



University
of Economics
in Katowice



your place



your space



your future

Climate risk in modern business

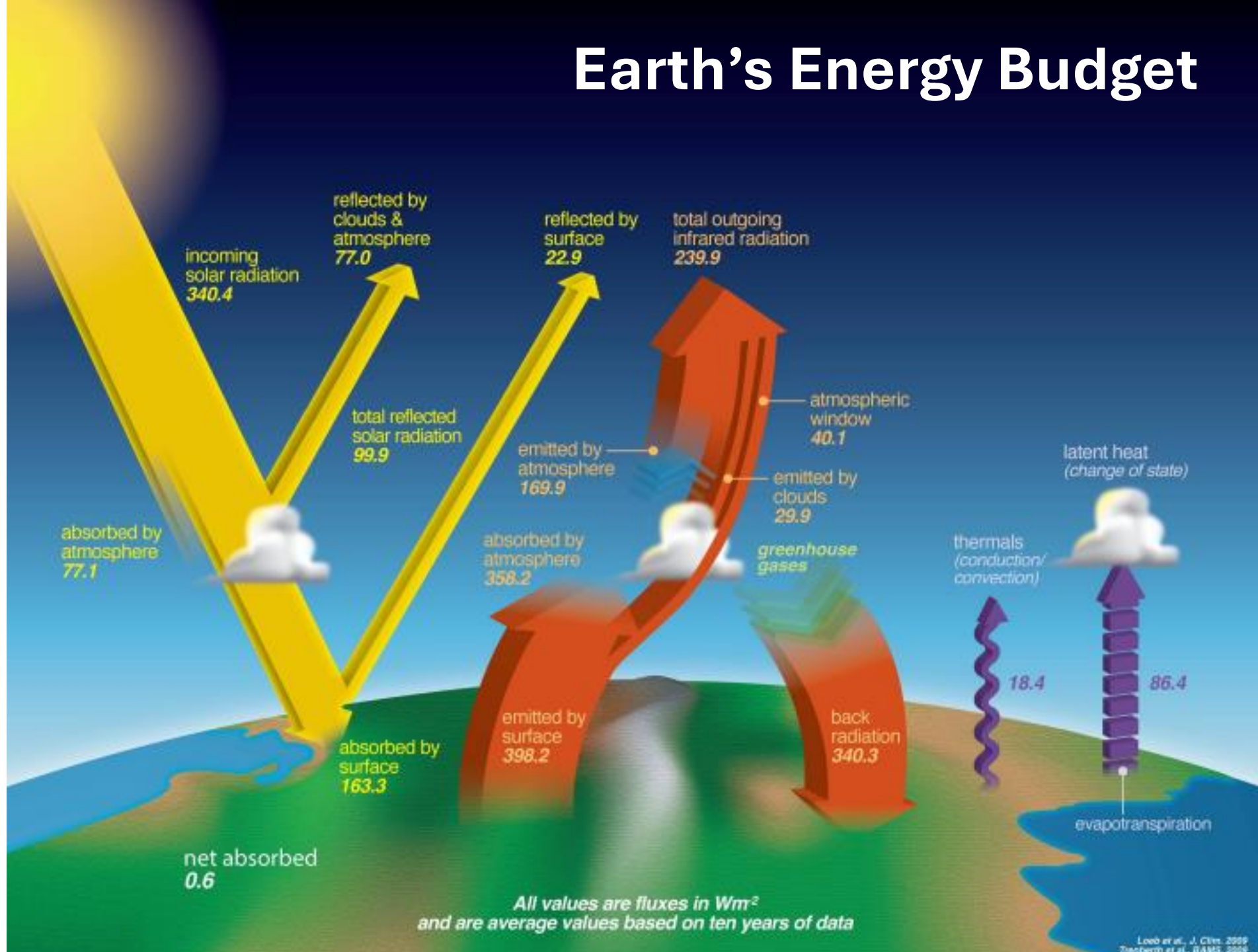
Ewa Dziwok

Department of Applied Mathematics

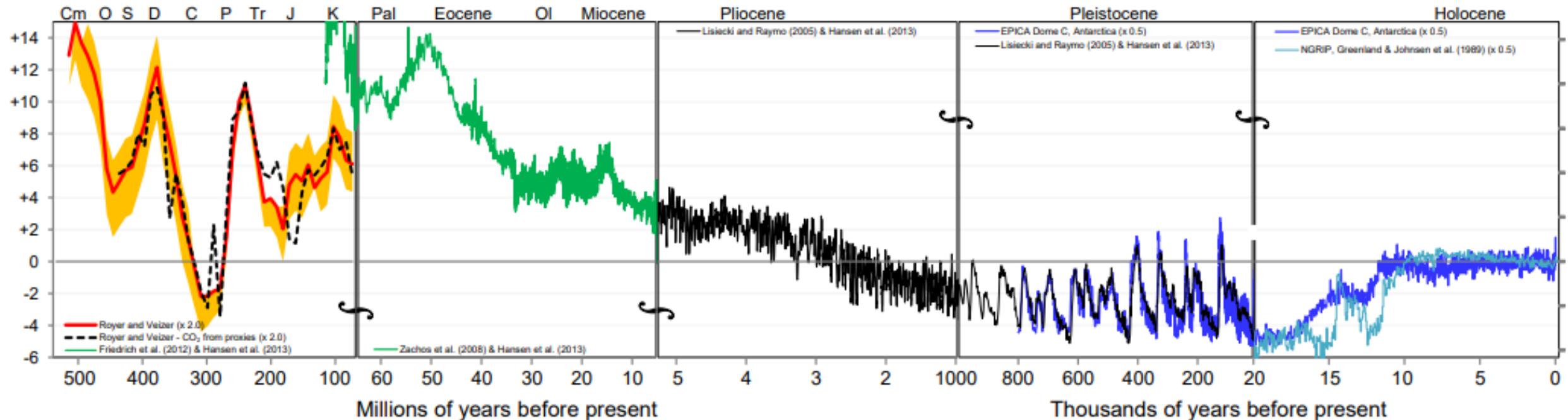
***SOME
THEORETICAL NOTES***

<https://www.youtube.com/watch?v=el7ygWztQqA>

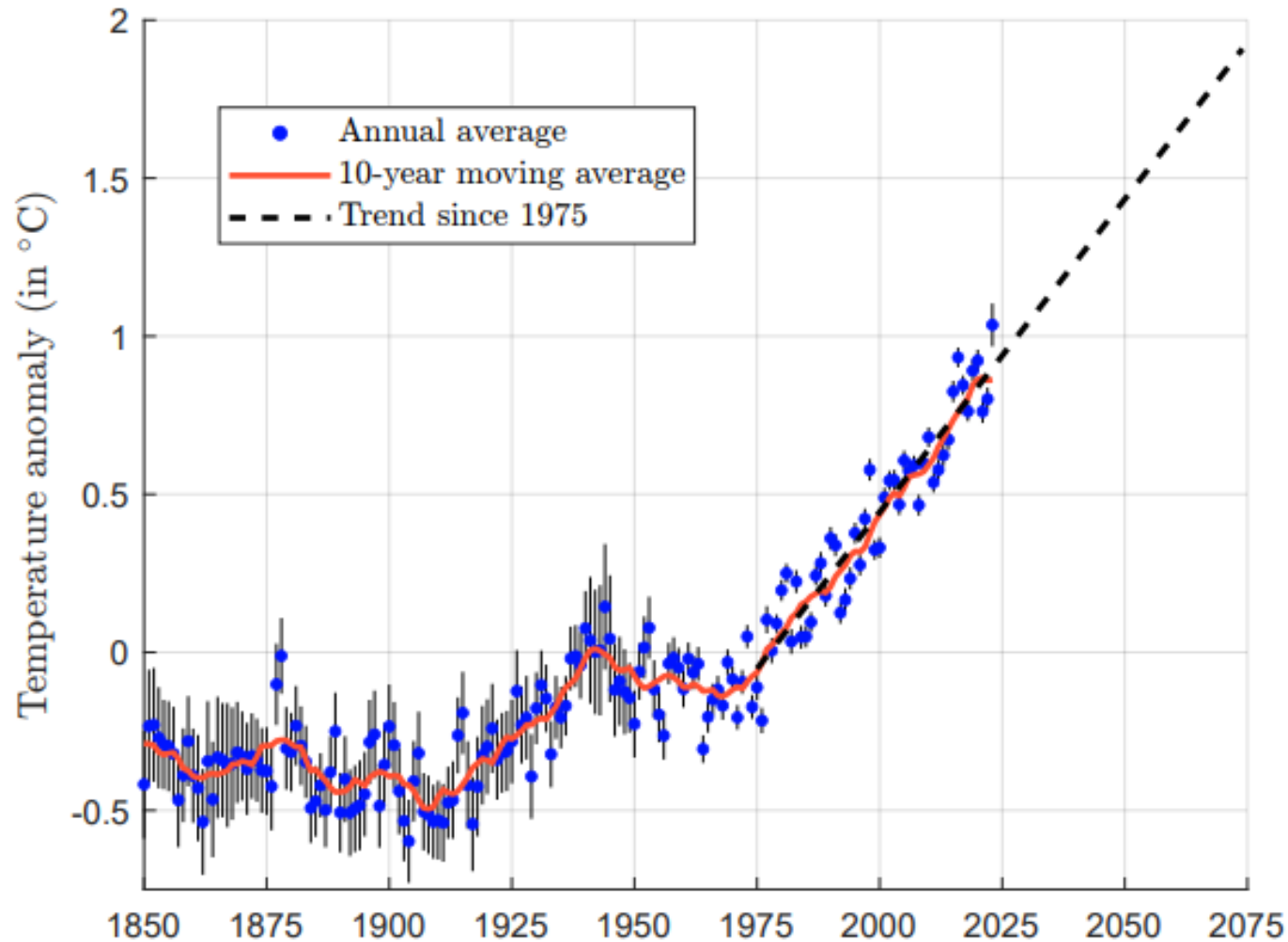
Earth's Energy Budget



Earth temperature since 500 Myr BP ($^{\circ}\text{C}$ vs. 1960-1990 average)

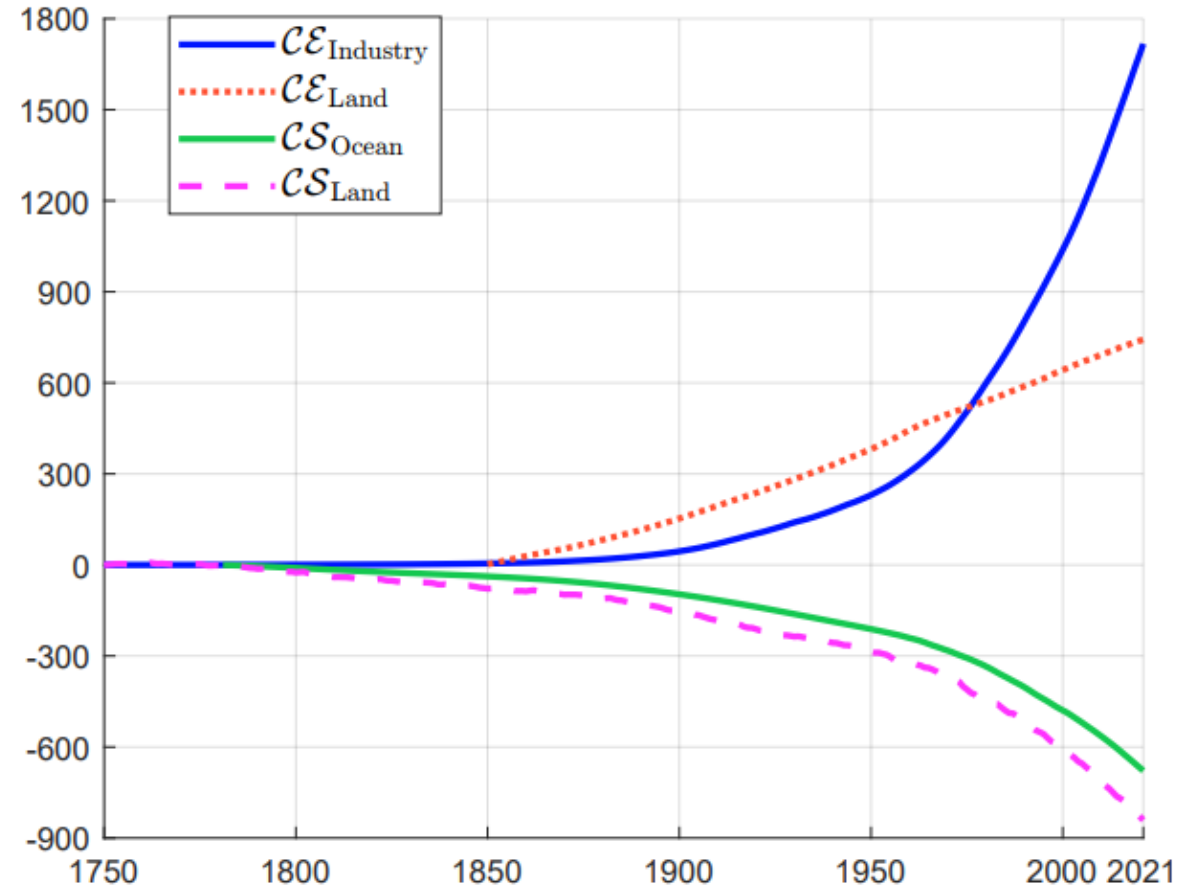
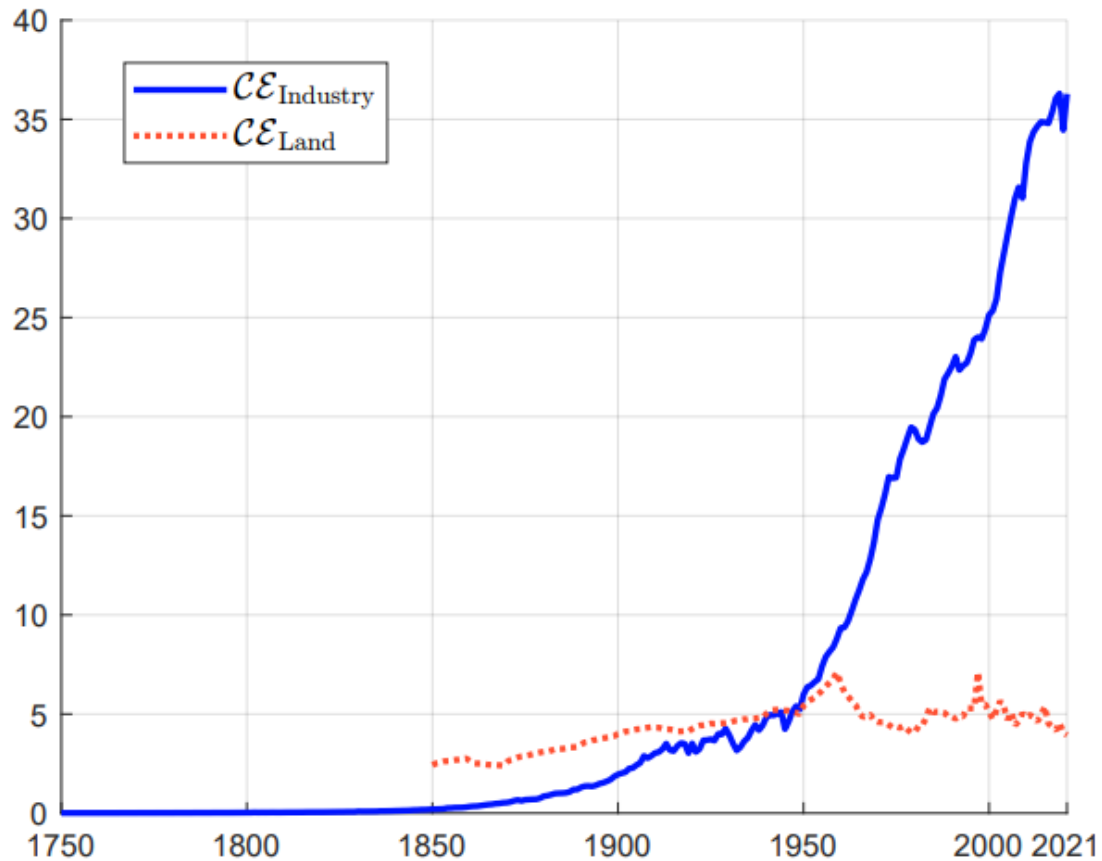


Global average land-ocean temperature anomaly relative to 1961-1990 average

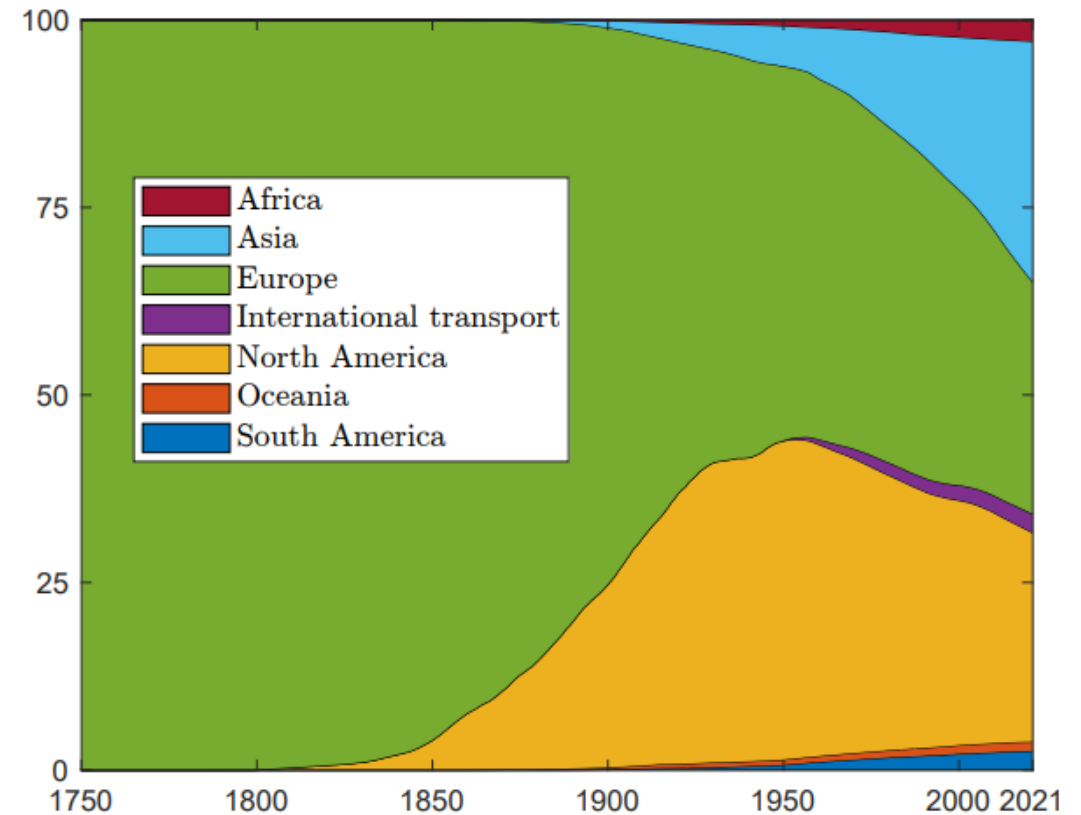
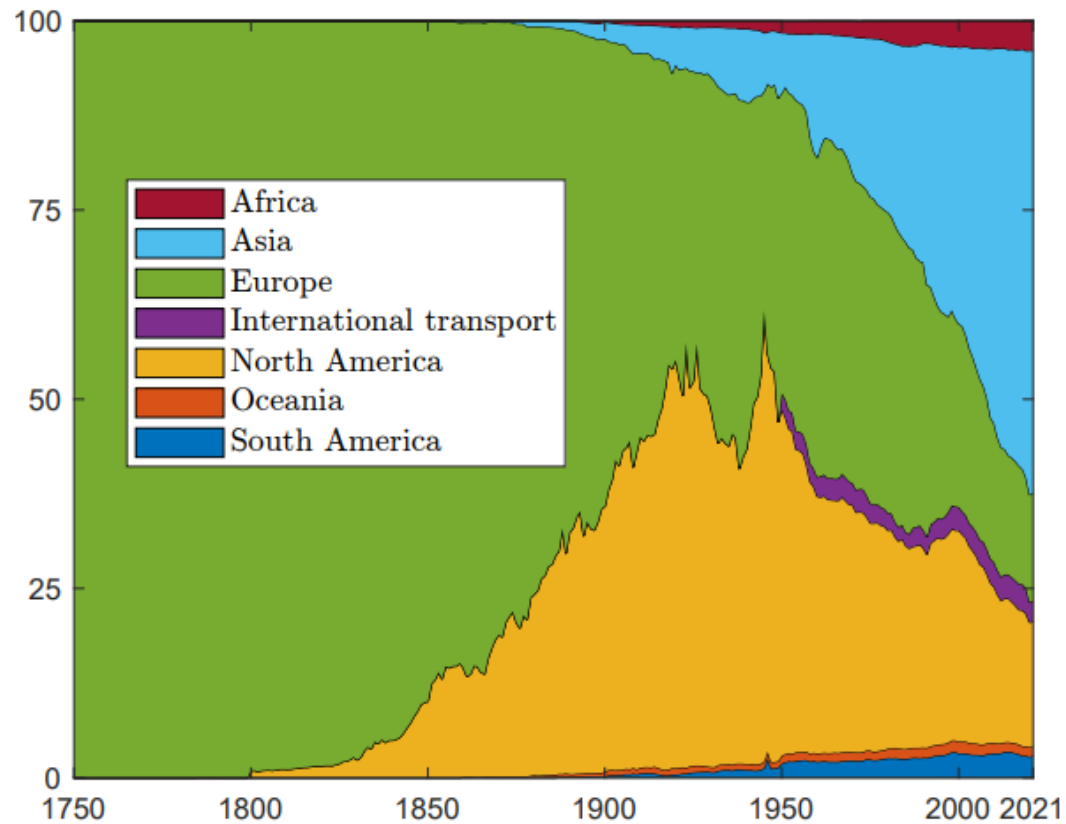


Source: Morice *et al.* (2021).

Annual CO₂ emissions / Cumulative CO₂ emissions and carbon sinks (in GtCO₂)

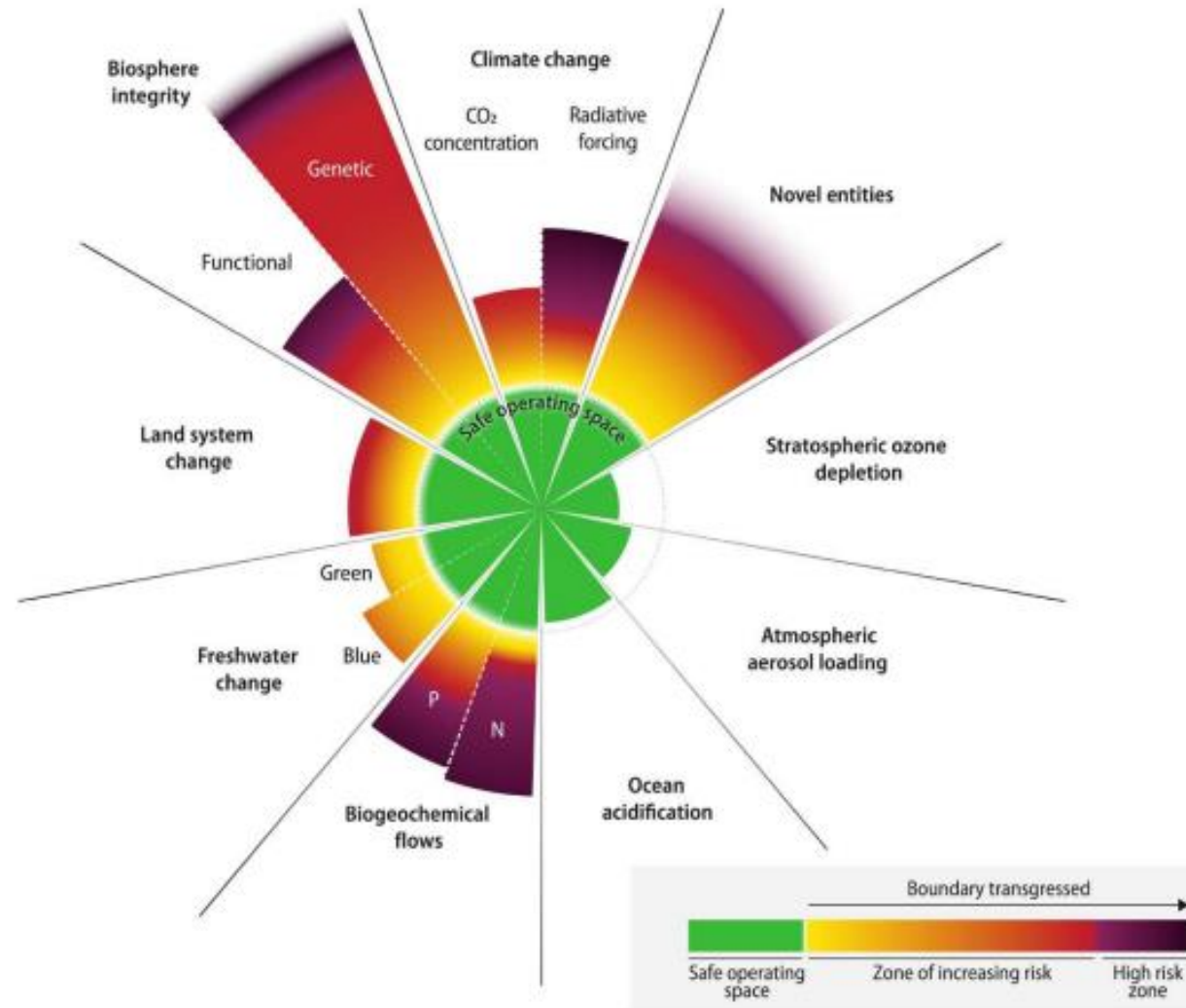


Share of CO₂ / cumulative CO₂ emissions by region (in % of total)



Source: Friedlingstein et al. (2022).

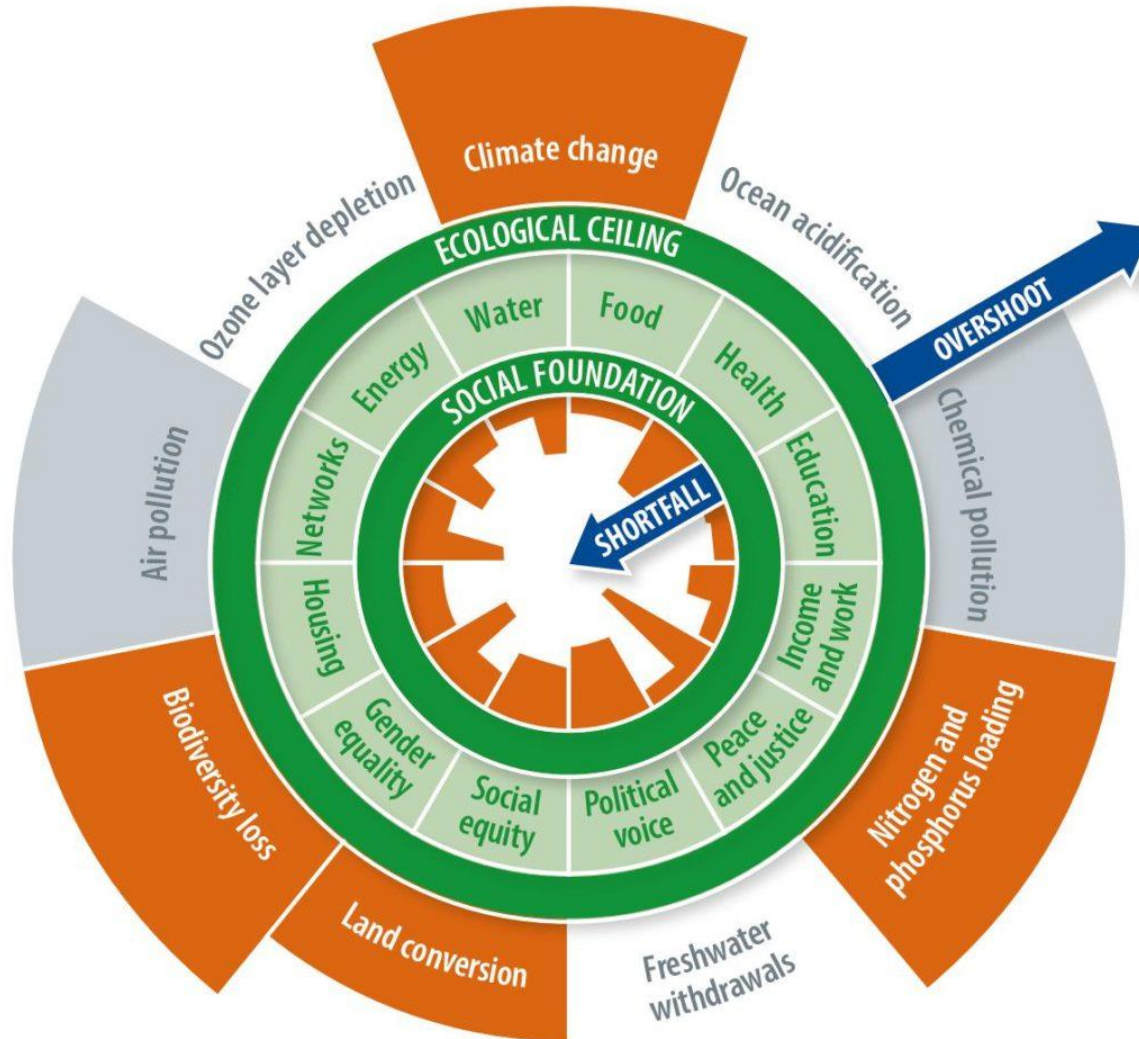
Planetary boundaries



Source: Richardson *et al.* (2023, Figure 1, page 4).

Doughnut (economic model)

The doughnut of social and planetary boundaries



<https://goodlife.leeds.ac.uk/>

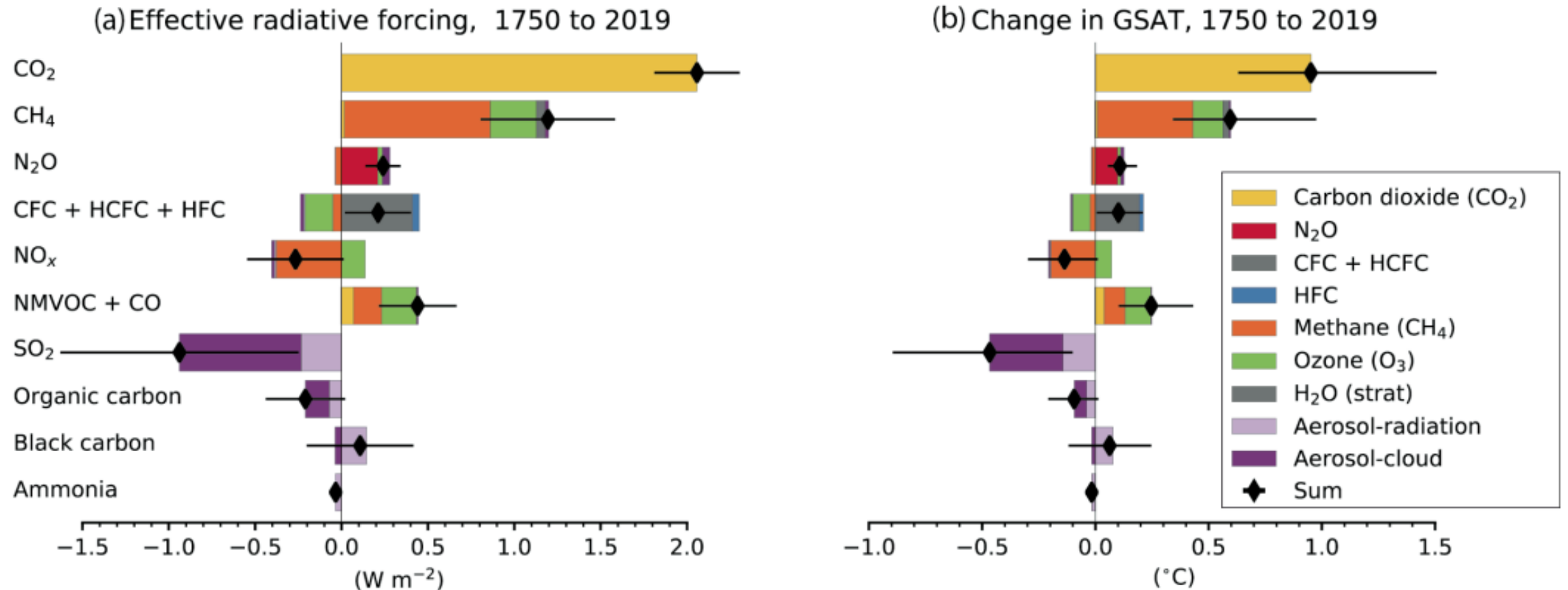
Climate sensitivity and feedback

- “[...] Climate forcing is a change to the climate system that can be expected to change the climate.
- A feedback process is a response of the climate system to surface warming that then alters the energy balance in such a way as to change the temperature response to the forcing.
- A **positive feedback** makes the forced response bigger, and a **negative feedback** makes it smaller. Classic examples of positive feedbacks are ice-albedo feedback and water-vapor feedback. When it warms, ice melts, and this reduces Earth’s albedo and causes further warming. When it cools, ice grows, and this increases Earth’s albedo, causing further cooling [...]” Hartmann (2016, page 294).

Contribution to effective radiative forcing (ERF)

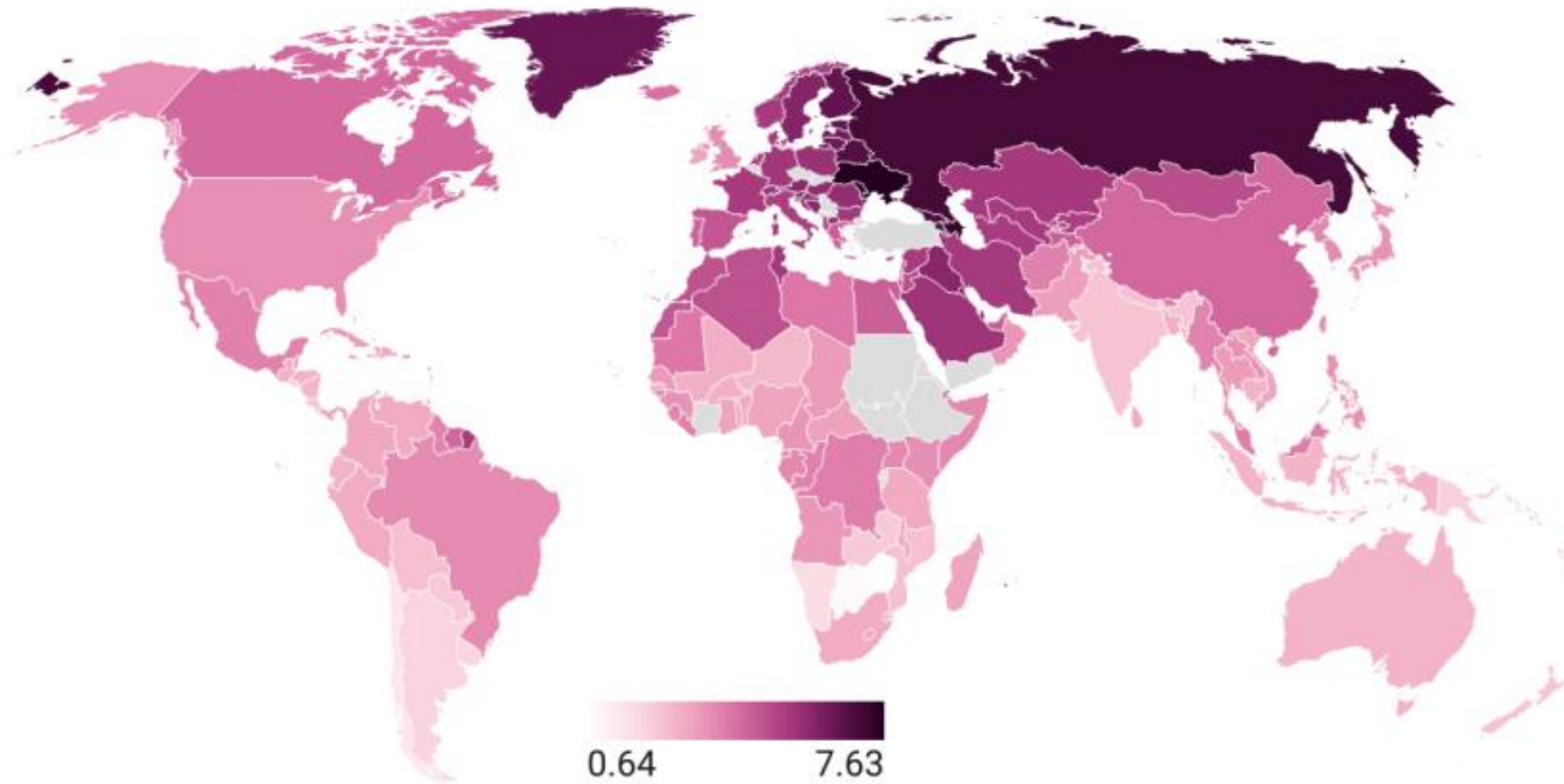
(a) and global mean surface air temperature (GSAT) change

(b) from component emissions between 1750 to 2019 based on CMIP6 models



Source: IPCC (2021, Figure 6.12, Chapter 6, page 854).

Projection of temperature anomaly by 2100 (in °C)



Source: <https://www.fao.org/faostat/en/#data/ET> & Author's calculations.

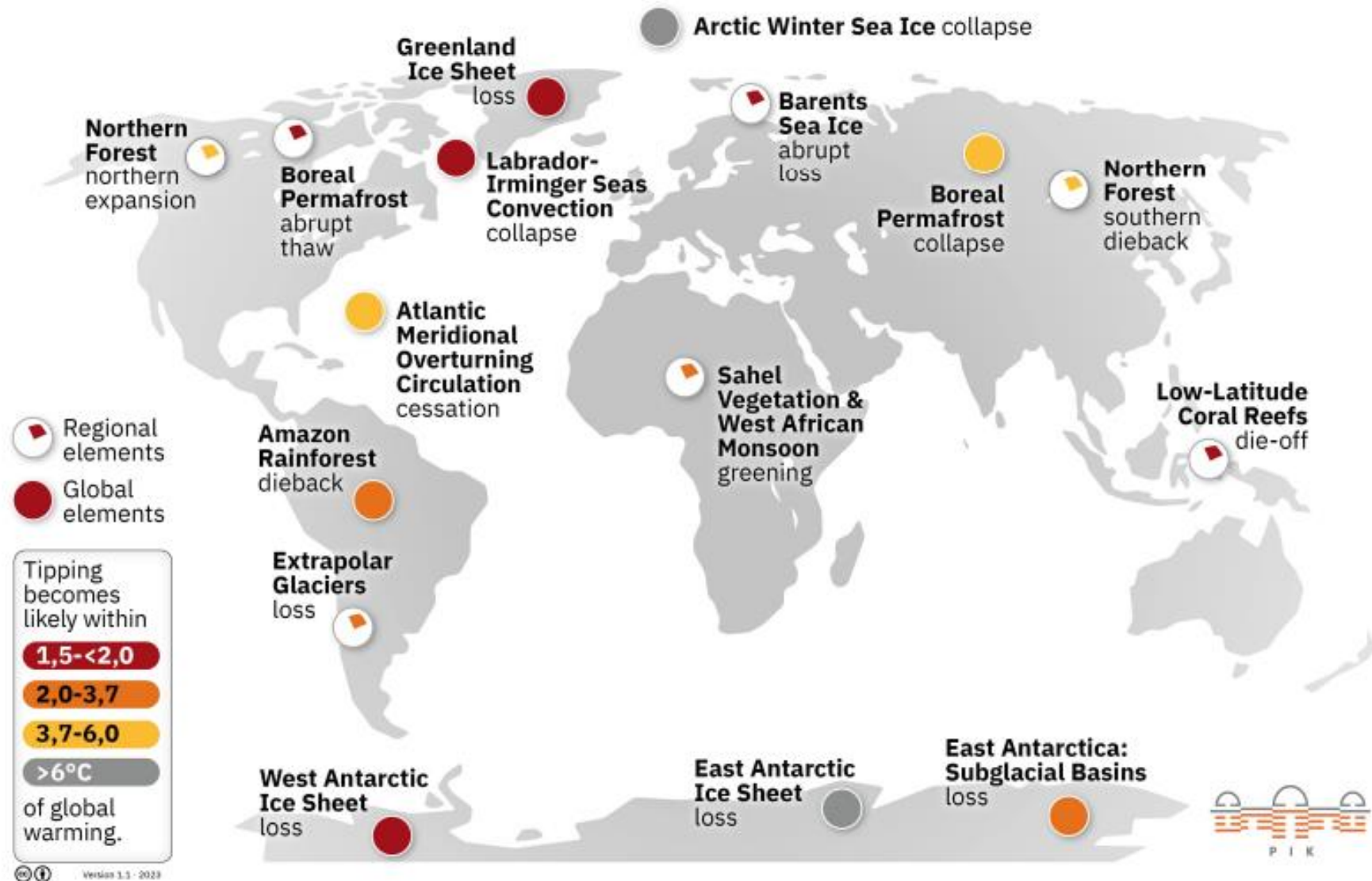
Tipping point definition

- The term was popularized in 2000 by Malcolm Gladwell in his book “The Tipping Point: How Little Things Can Make a Big Difference”, which explored the concept in sociological change
- In climate change, it was popularized in the late 2000s by Lenton et al. (2008)
- In common parlance, a tipping point is a change:
“A climate tipping point occurs when a small change in forcing triggers a strongly nonlinear response in the internal dynamics of part of the climate system, qualitatively changing its future state.”
(Lenton et al., 2011, page 201).

Tipping point definition

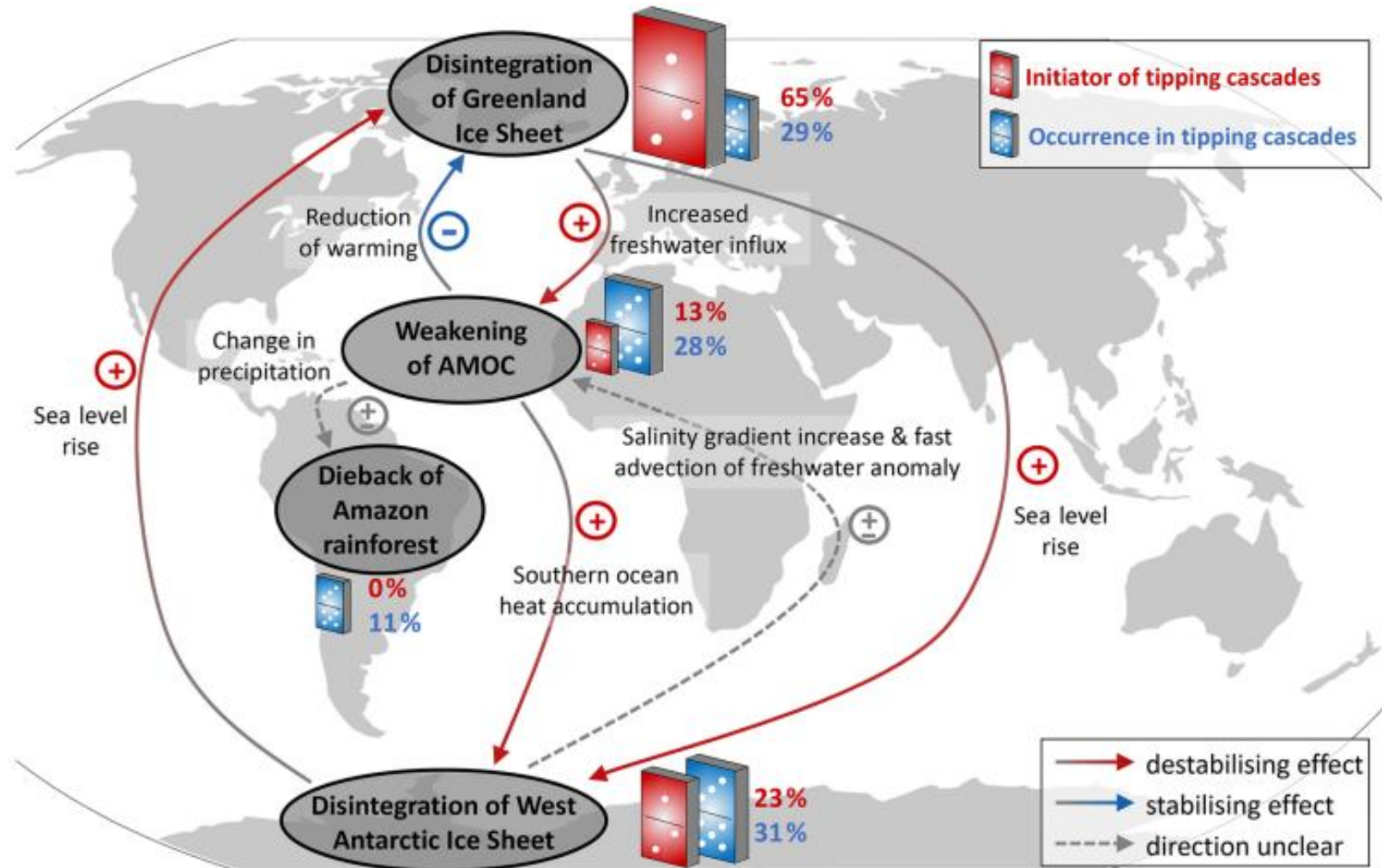
- “Tipping points refer to **critical thresholds** in a system that, when exceeded, can lead to a significant change in the state of the system, often with an understanding that the change is irreversible.” (IPCC, 2018, page 262).
- A tipping point is “a hypothesized critical threshold when global or regional climate changes from one stable state to another stable state. The tipping point event may be **irreversible**.” (IPCC, 2021, page 1463).

Geographical distribution of global and regional tipping elements



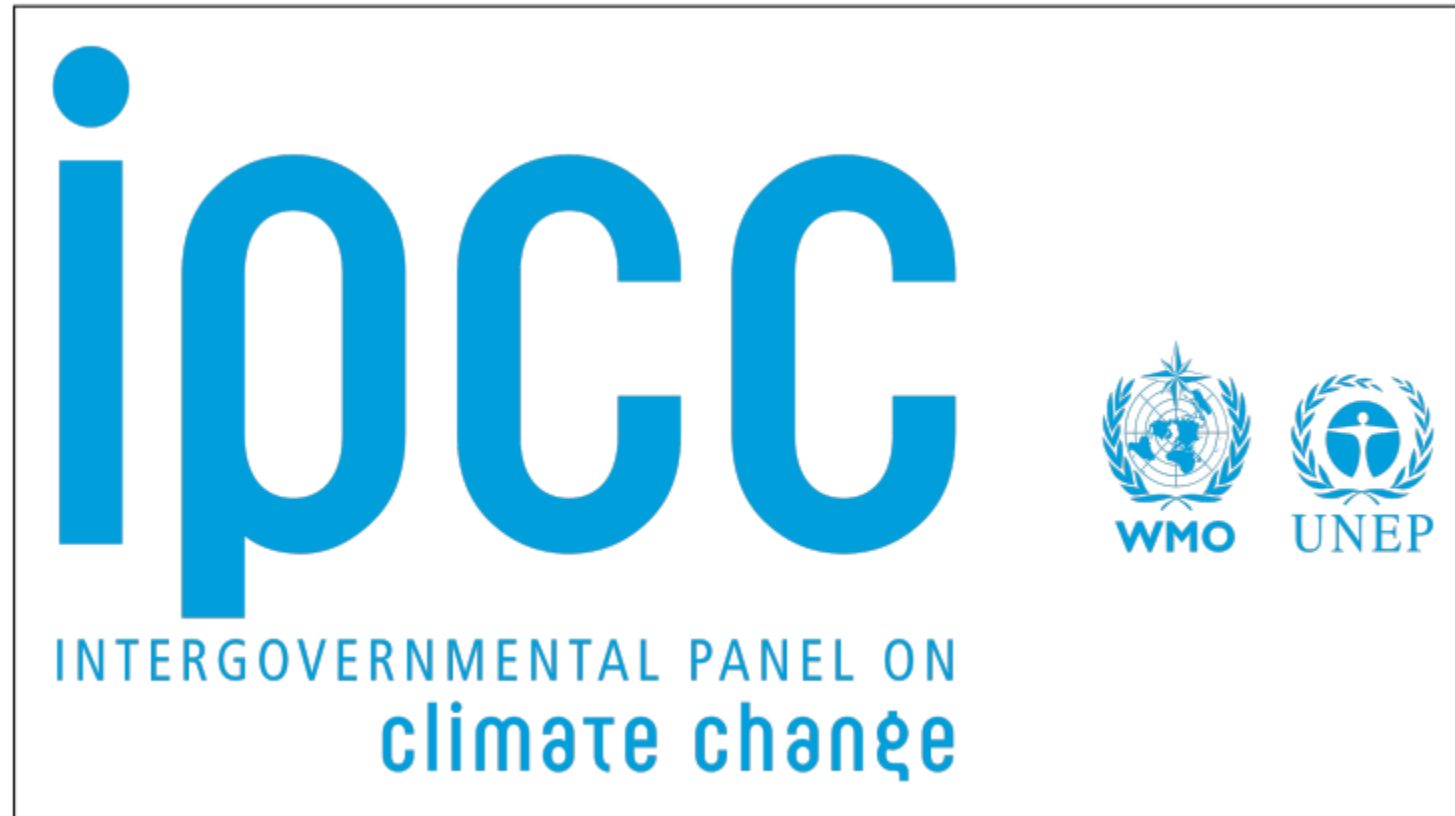
Source: Armstrong McKay *et al.* (2022).

Interactions between climate tipping elements and their roles in tipping cascades - domino effects



Source: Wunderling *et al.* (2021, Figure 1, page 603).

Intergovernmental Panel on Climate Change



Intergovernmental Panel on Climate Change

- The United Nations Environment Programme (UNEP) and the World Meteorological Organization (WMO) established the Intergovernmental Panel on Climate Change in 1988
- The members of the IPCC are then governments (95 as of January 2024), not scientists, academic institutions, or NGOs
- Its original mandate was to “*prepare a comprehensive review and recommendations on the state of knowledge of the science of climate change, the social and economic impacts of climate change, and possible response strategies and elements for inclusion in a possible future international climate convention*”

Intergovernmental Panel on Climate Change

- **Working Group I (WGI)** Assesses the scientific aspects of the climate system that underpin past, present and future climate change.
- **Working Group II (WGII)** Assesses climate change impacts, adaptation options and vulnerabilities on human and natural systems.
- **Working Group III (WGIII)** Focuses on mitigation of climate change, assessment of methods to limit and reduce greenhouse gas emissions, and removal of greenhouse gases from the atmosphere.

List of IPCC Assessment Reports

Date	Symbol	Title of the report
1990	FAR/AR1	First IPCC Assessment Report
1995	SAR/AR2	Second Assessment Report
2001	TAR/AR3	Third Assessment Report
2007	AR4	Fourth Assessment Report
2014	AR5	Fifth Assessment Report
2023	AR6	Sixth Assessment Report

The three main conclusions of the AR6 Synthesis Report are:

- 1. Unequivocal human influence** The report reaffirms that human activities are unequivocally causing climate change. The global surface temperatures have reached 1.1 °C above pre-industrial levels.
- 2. Widespread and intensifying impacts** The report emphasizes that climate change is already having a profound impact on the planet, affecting ecosystems, human health, and infrastructure. These impacts are projected to worsen in the future, even with moderate warming.
- 3. Urgency of mitigation and adaptation** The report underscores the urgency of taking action to address climate change. Limiting global warming to 1.5 °C will require rapid and far-reaching changes in energy systems, food production and land management. These changes are possible and would bring significant environmental and human health benefits.

Earth Summit(s)

- **June 1972:** United Nations Conference on the Human Environment or [Stockholm conference](#)
 - ✓ Creation of the United Nations Environment Programme (UNEP)
- **June 1992:** United Nations Conference on Environment and Development (UNCED) or the Earth Summit ([Rio de Janeiro](#), Brazil)
 - ✓ Rio Declaration on Environment and Development signed by 170 countries
 - ✓ Forest Principles (also known as the Rio Forest Principles)

Earth Summit(s)

- **Agenda 21** ⇒ Sustainable Development Goals (SDGs)
 - ✓ Social and economic dimensions.
 - ✓ Conservation and management of resources for development
 - ✓ Strengthening the role of major groups
 - ✓ Means of implementation
- The so-called **Rio Conventions**
 - ✓ UN Framework Convention on Climate Change (UNFCCC)
 - ✓ Convention on Biological Diversity (CBD)
 - ✓ United Nations Convention to Combat Desertification (UNCCD)
- Launch of the **UNEP Finance Initiative** (UNEP-FI)

Conference of the Parties (COP)

Year	COP	CMP	CMA	City	Country	Treaty
1992				Rio de Janero	Brazil	Convention
1995	1			Berlin	Germany	
1996	2			Geneva	Switzerland	
1997	3			Kyoto	Japan	Kyoto Protocol
1998	4			Buenos Aires	Argentina	
1999	5			Bonn	Germany	
2000	6-1			The Hague	Netherlands	
2001	6-2			Bonn	Germany	
2001	7			Marrakech	Morocco	
2002	8			New Delhi	India	
2003	9			Milan	Italy	
2004	10			Buenos Aires	Argentina	
2005	11	1		Montreal	Canada	
2006	12	2		Nairobi	Kenya	
2007	13	3		Bali	Indonesia	
2008	14	4		Poznań	Poland	
2009	15	5		Copenhagen	Denmark	

Conference of the Parties (COP)

Year	COP	CMP	CMA	City	Country	Treaty
2010	16	6		Cancún	Mexico	
2011	17	7		Durban	South Africa	
2012	18	8		Doha	Qatar	
2013	19	9		Warsaw	Poland	
2014	20	10		Lima	Peru	
2015	21	11		Paris	France	Paris Agreement
2016	22	12	1-1	Marrakech	Morocco	
2017	23	13	1-2	Bonn	Germany	
2018	24	14	1-3	Katowice	Poland	
2019	25	15	2	Madrid	Spain	
2021	26	16	3	Glasgow	United Kingdom	
2022	27	17	4	Sharm El Sheikh	Egypt	
2023	28	18	5	Dubai	United Arab Emirates	
2024	29	19	6	Baku	Azerbaijan	

Important COPs

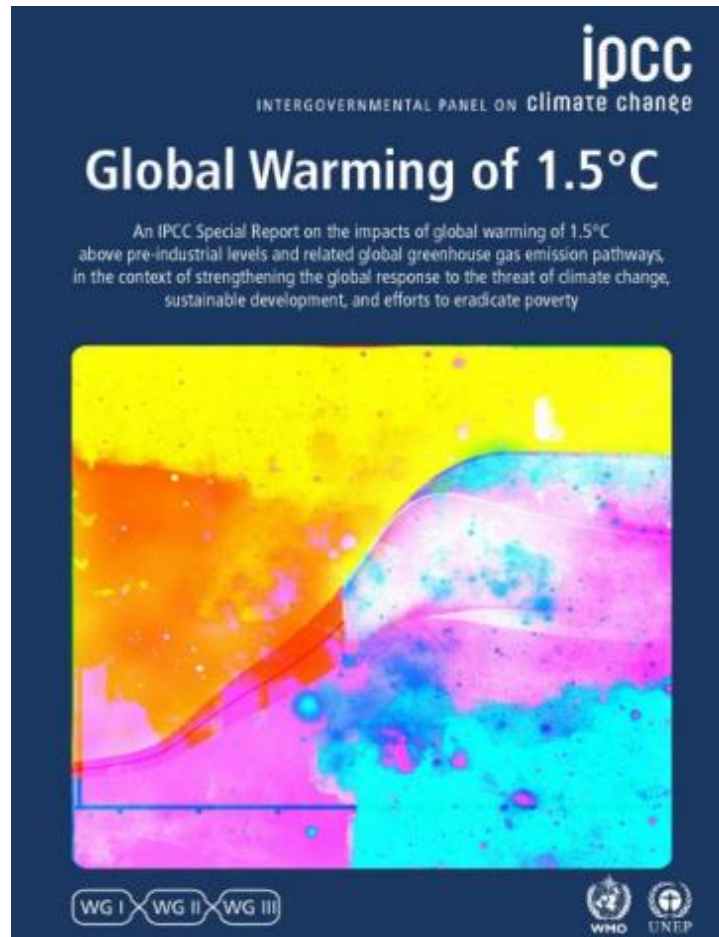
- COP1: *Berlin* (1995)
- COP3: *Kyoto* (1997) $\Rightarrow$ Kyoto Protocol (CMP) (entered 16 Feb 2005)
- COP15: *Copenhagen* (2009) $\Rightarrow$ 2°C limit
- COP18: *Doha* (2012) $\Rightarrow$ Second commitment of the Kyoto Protocol
- COP21: *Paris* (2015) $\Rightarrow$ Paris Agreement (CMA)
- COP26: *Glasgow* (2021) $\Rightarrow$ GFANZ & Net Zero
- COP28: *Dubai* (2023) $\Rightarrow$ “Beginning of the end” of the fossil fuel era

Paris Agreement

The Paris Agreement is an international treaty with the following goals:

- 1) Keep a global temperature rise this century well below 2° C above the pre-industrial levels
 - 2) Pursue efforts to limit the temperature increase to 1.5 ° C
 - 3) Increase the ability of countries to deal with the impacts of climate change
 - 4) Make finance flows consistent with low GHG emissions and climate-resilient pathways
- ⇒ Nationally determined contribution (NDC)

IPCC Special Report on Global Warming of 1.5 °C (SR15) / The Economist, Say goodbye to 1.5 °C, November 5, 2022

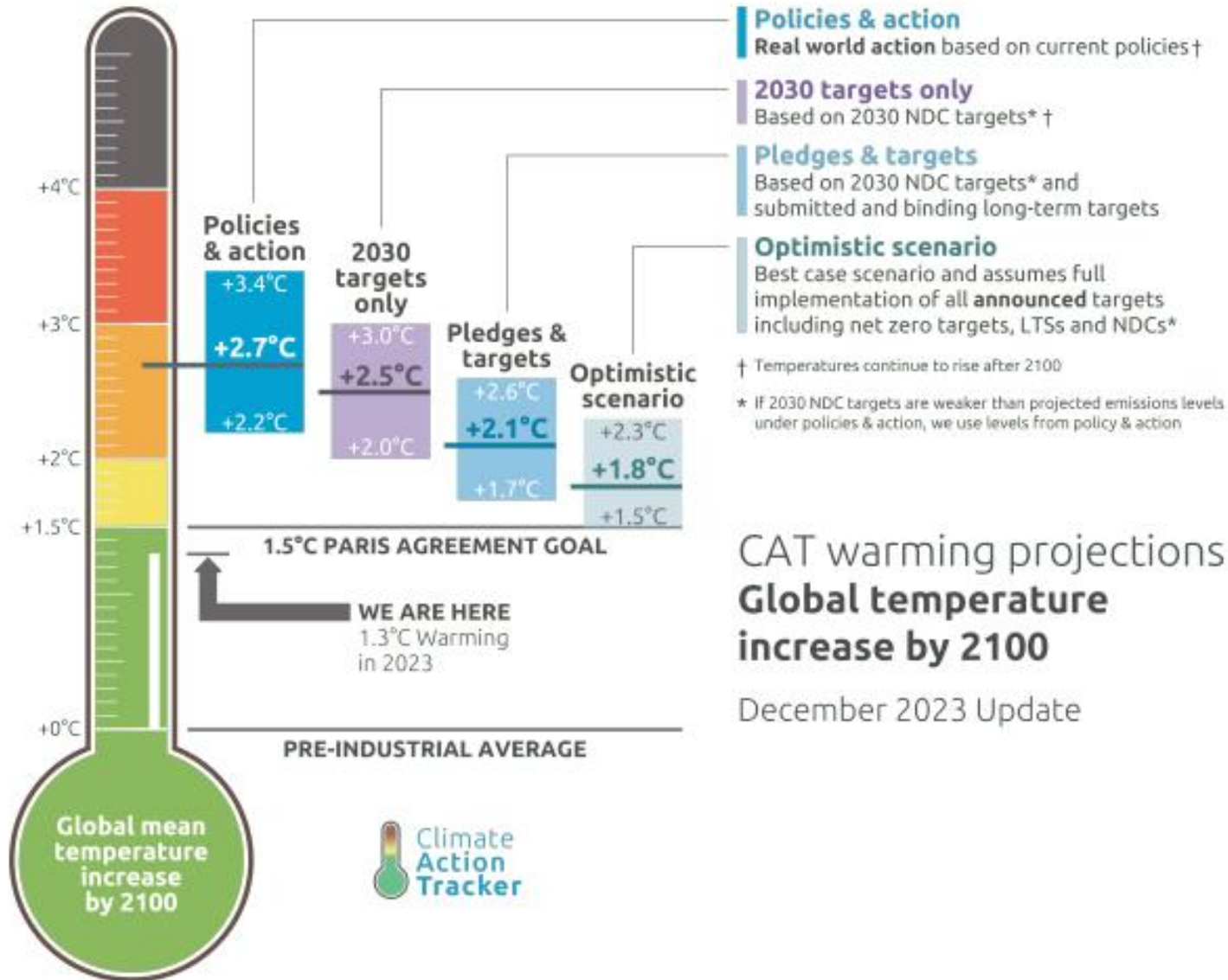


2022 GHG emissions and NDCs of top emitting countries

Rank	Country	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
1	China	15 685	29.16	29.16	10.95	0.61	35.62	65	2005	2030	CIR
2	United States	6 017	11.19	40.35	17.90	0.28	−10.01	50	2005	2030	AER
3	India	3 943	7.33	47.68	2.79	0.39	38.85	45	2005	2030	CIR
4	Russia	2 580	4.80	52.48	17.99	0.64	14.90	30	1990	2030	AER
5	Brazil	1 310	2.44	54.91	6.05	0.40	11.27	50	2005	2030	AER
6	Indonesia	1 241	2.31	57.22	4.47	0.36	50.00	32	BAU	2030	RER
7	Japan	1 183	2.20	59.42	9.41	0.23	−10.85	46	2013	2030	AER
8	Iran	952	1.77	61.19	11.20	0.70	17.94				
9	Mexico	820	1.52	62.71	5.99	0.33	6.64	22	BAU	2030	RER
10	Saudi Arabia	811	1.51	64.22	22.64	0.45	30.84	34	2020	2030	RER

(1) GHG emissions in 2022 in MtCO₂e, (2) share in %, (3) cumulative share in%, (4) GHG emissions per capita in 2022 in tCO₂e, (5) GHG emissions per GDP in 2022 in kgCO₂e, (6) GHG emissions growth between 2010 and 2022 in %, (7) NDC reduction rate, (8) base year (BAU = business-as-usual), (9) target year, (10) mitigation type (AER = absolute emissions reduction, CIR = carbon intensity reduction, RER = relative emissions reduction).

The CAT thermometer



CAT warming projections
**Global temperature
increase by 2100**

December 2023 Update

European Green Deal (December 2019)

Adopted by the EU Council in June 2021, the European Climate Law includes the following components:

- Set a 2030 target to reduce net greenhouse gas emissions by at least 55% compared to 1990 levels;
- Establish a system to monitor progress and take further action if necessary;
- Establish the [European Scientific Advisory Board on Climate Change](#) to provide independent scientific advice;
- Define the EU-wide greenhouse gas emission reduction path for 2030-2050;
- Provide predictability for investors and other economic actors;
- Ensure that the transition to climate neutrality is irreversible;
- Commit to [negative emissions after 2050](#).

Communication on the **European Green Deal**, 14 July 2021

- In July 2021, the European Commission presented the **Fit for 55** package
- First quantitative target of the **European Climate Change Law**, namely, to reduce Europe's net greenhouse gas emissions by at least 55% by 2030
- The package consists of several legislative proposals
- As of 1 January 2024, 12 proposals have been adopted

Fit for 55 package

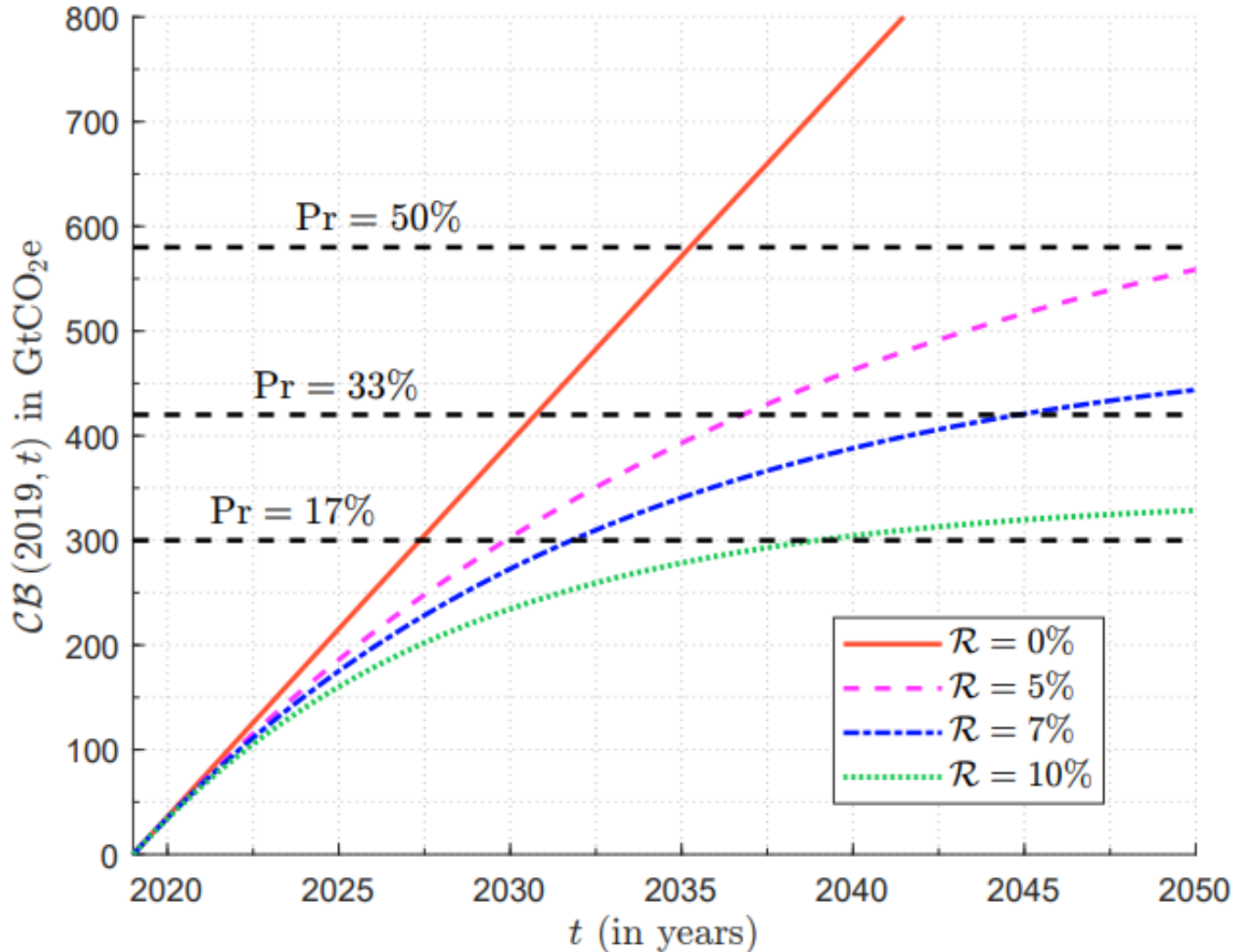
1. Reform of the emissions trading system (ETS, April 2023)
2. New emissions trading system for building and road transport fuels (April 2023)
3. Social climate fund (April 2023)
4. Revision of the effort sharing regulation (ESR, March 2023)
5. Revision of the regulation on land use, forestry and agriculture (LULUCF, March 2023)
6. Revised regulation on CO2 emissions from new cars and vans (March 2023)
7. Carbon border adjustment mechanism (CBAM, April 2023)
8. Renewable energy directive (RED III, October 2023)
9. Energy efficiency directive (EED, July 2023)
10. Alternative fuels infrastructure regulation (AFIR)
11. ReFuelEU aviation regulation
12. FuelEU maritime regulation

National reduction targets under the ESR (revised 2023 vs. original 2018)

Country	2018	2023	Country	2018	2023	Country	2018	2023
Austria	36.0%	48.0%	France	37.0%	47.5%	Malta	19.0%	19.0%
Belgium	35.0%	47.0%	Germany	38.0%	50.0%	Netherlands	36.0%	48.0%
Bulgaria	0.0%	10.0%	Greece	16.0%	22.7%	Poland	7.0%	17.7%
Croatia	7.0%	16.7%	Hungary	7.0%	17.7%	Portugal	17.0%	28.7%
Cyprus	24.0%	32.0%	Ireland	30.0%	42.0%	Romania	2.0%	12.7%
Czechia	14.0%	26.0%	Italy	33.0%	43.7%	Slovakia	12.0%	22.7%
Denmark	39.0%	50.0%	Latvia	6.0%	17.0%	Slovenia	15.0%	27.0%
Estonia	13.0%	24.0%	Lithuania	9.0%	21.0%	Spain	26.0%	37.7%
Finland	39.0%	50.0%	Luxembourg	40.0%	50.0%	Sweden	40.0%	50.0%

Source: European Commission (2023), <https://climate.ec.europa.eu/eu-action/effort-sharing-member-states-emission-targets>.

Carbon budget and global warming / Probability to reach 1.5 °C



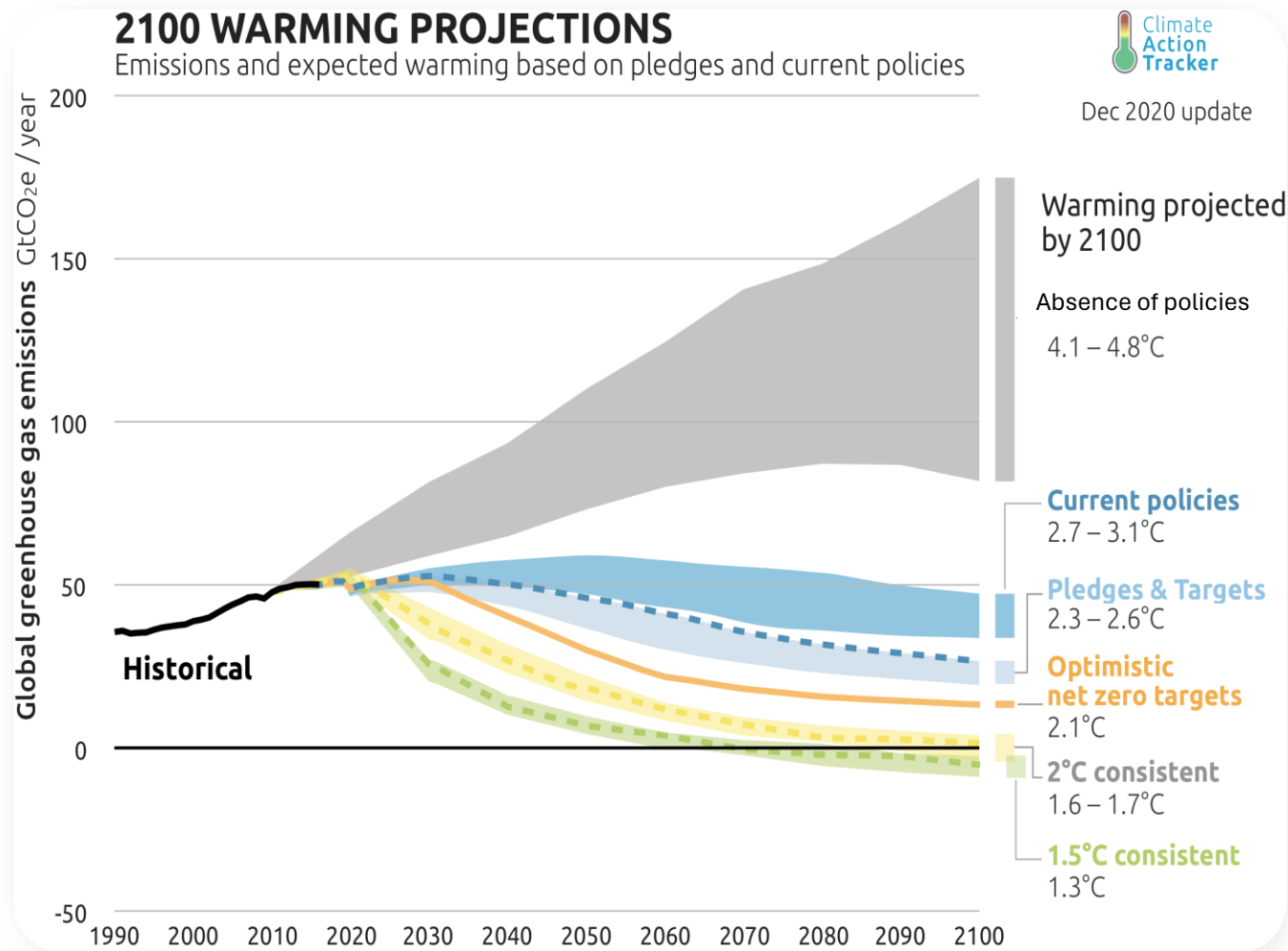
IPCC (2018)

The remaining carbon budget $CB(2019, t)$ is:

- 580 GtCO₂e for a 50% probability of limiting warming to 1.5°C
- 420 GtCO₂e for a 66% probability
- 300 GtCO₂e for a 83% probability

CLIMATE RISK
(1) PHYSICAL RISK

Climate change is leading to physical and transition risks



Physical risks – arising from climate change (worse storms, more wildfires, rising sea level)

Transition risks – arising from the adjustment towards a low-carbon economy (e.g. policy, legal, technology changes, reputational risks)

Physical risks from a changing climate

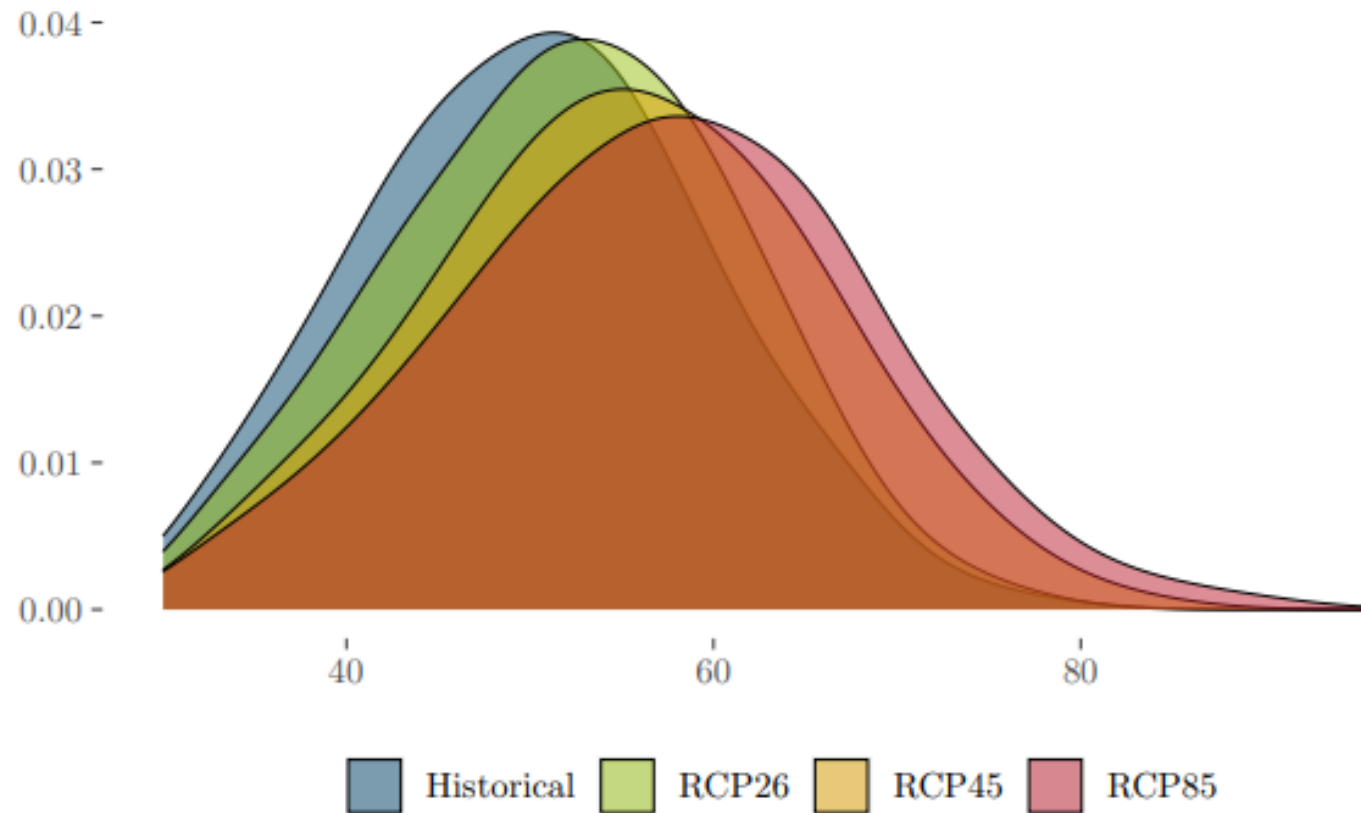
Acute

- **Storms:** Record-breaking 30 Atlantic named storms in 2020. Costliest hurricane season ever in 2017
- **Floods:** FEMA in 2020 noted, “It takes 7 years on average to complete a new flood map, meaning that some maps may technically be out of date by the time they are finished”
- **Heatwave:** The 2003 heatwave in Northern Europe, with 70,000 deaths, a 1in500 return period; models predict this will be 1in2 by 2040s

Chronic

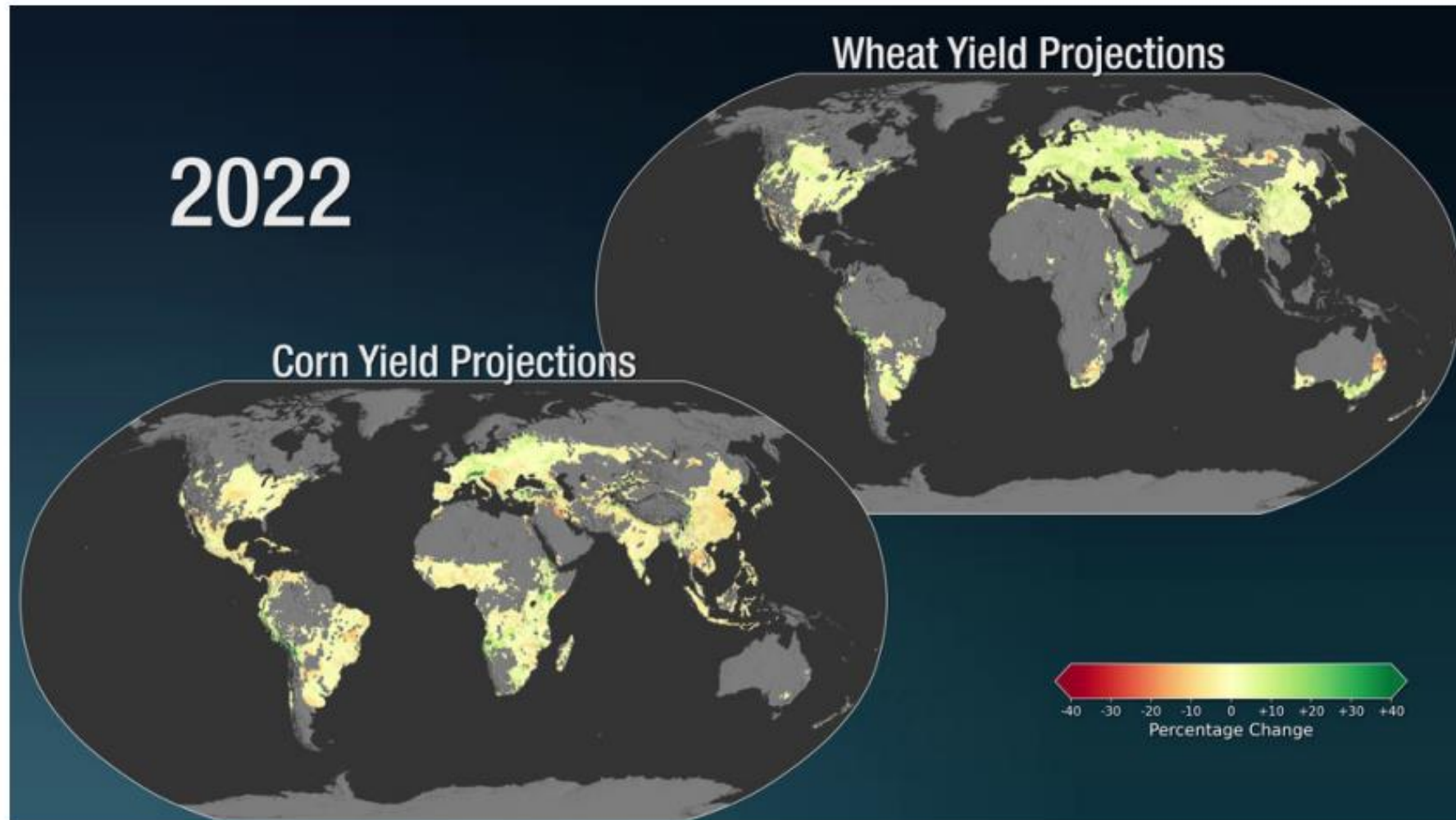
- **Sea-level rise:** At 2.0°C global average sea level by 2100 will be 0.46m – major investment needed in ports
- **Temperature & precipitation:** In 2009 the estimated annual cost of corrosion worldwide was \$1.8 trillion, i.e. 3%-4% of industrialised countries' GDP. These costs will rise
- **Sea temperature:** 99% of all coral reefs will be lost at 2.0°C
- **Heat stress:** 38°C (100°F) will be normal summer temperatures in Southern England by 2040

Maximum wind speed in m/s (2070-2100)



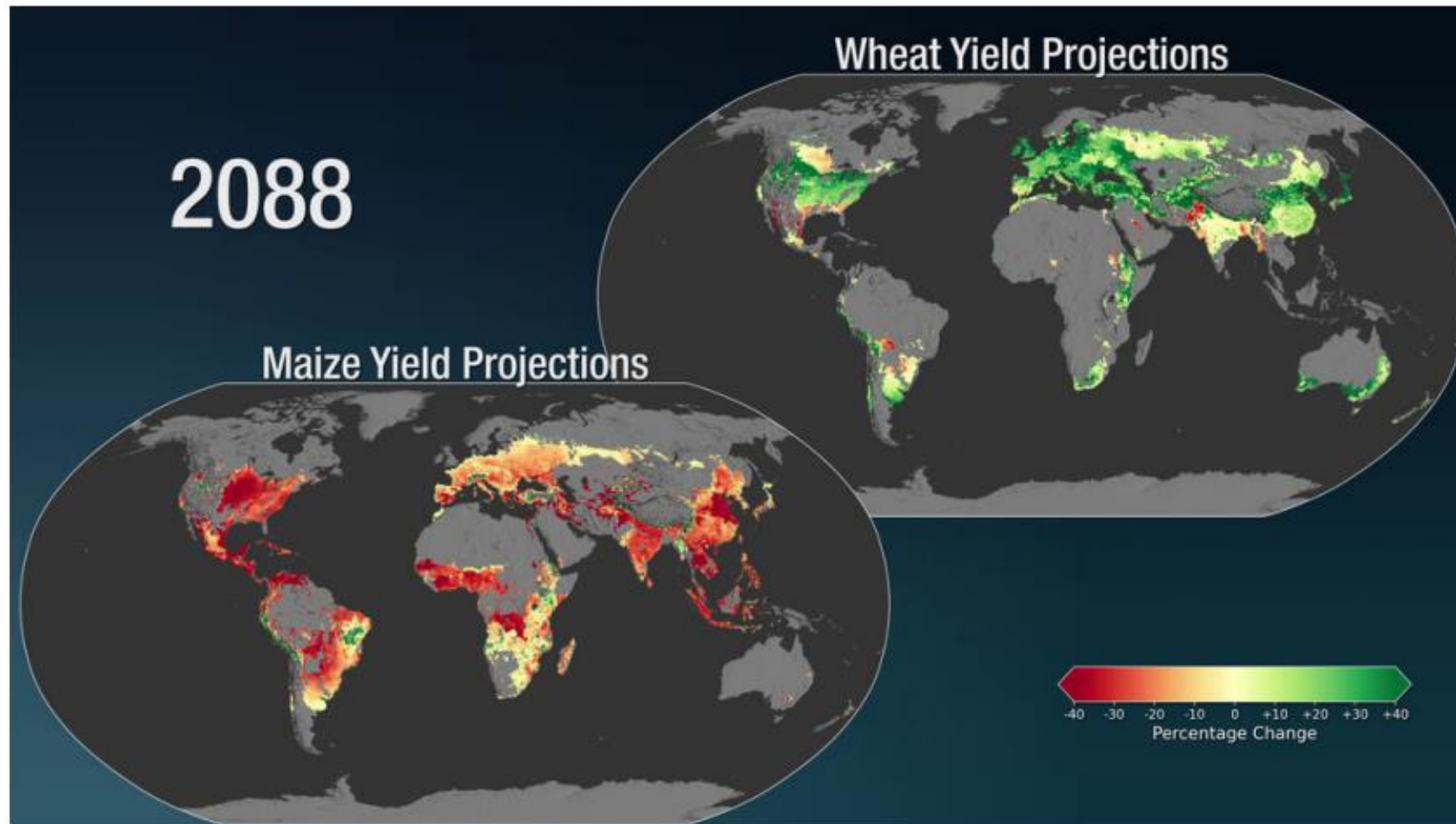
Source: Le Guenedal *et al.* (2021).

Crop yield* (2088 vs. 2022)



Source: Jägermeyr *et al.* (2021) & <https://svs.gsfc.nasa.gov/4974>.

Crop yield* (2088 vs. 2022)



Source: Jägermeyr *et al.* (2021) & <https://svs.gsfc.nasa.gov/4974>.

CLIMATE RISK
(2) TRANSITION RISK

Climate transition risk

The carbon footprint approach assumes that the climate-related market risk of a company is measured by its current carbon intensity

...But the market perception of the climate change may be different

Carbon emissions



Scope 1 – all direct emissions from the activities of an organisation under their control (e.g., fuel combustion on site, from owned vehicles)



Scope 2 – indirect emissions from electricity purchased and used by the firm



Scope 3 – all other indirect emissions which arise from sources a firm does not control (e.g., from business travel, procurement, waste and water)

Carbon price

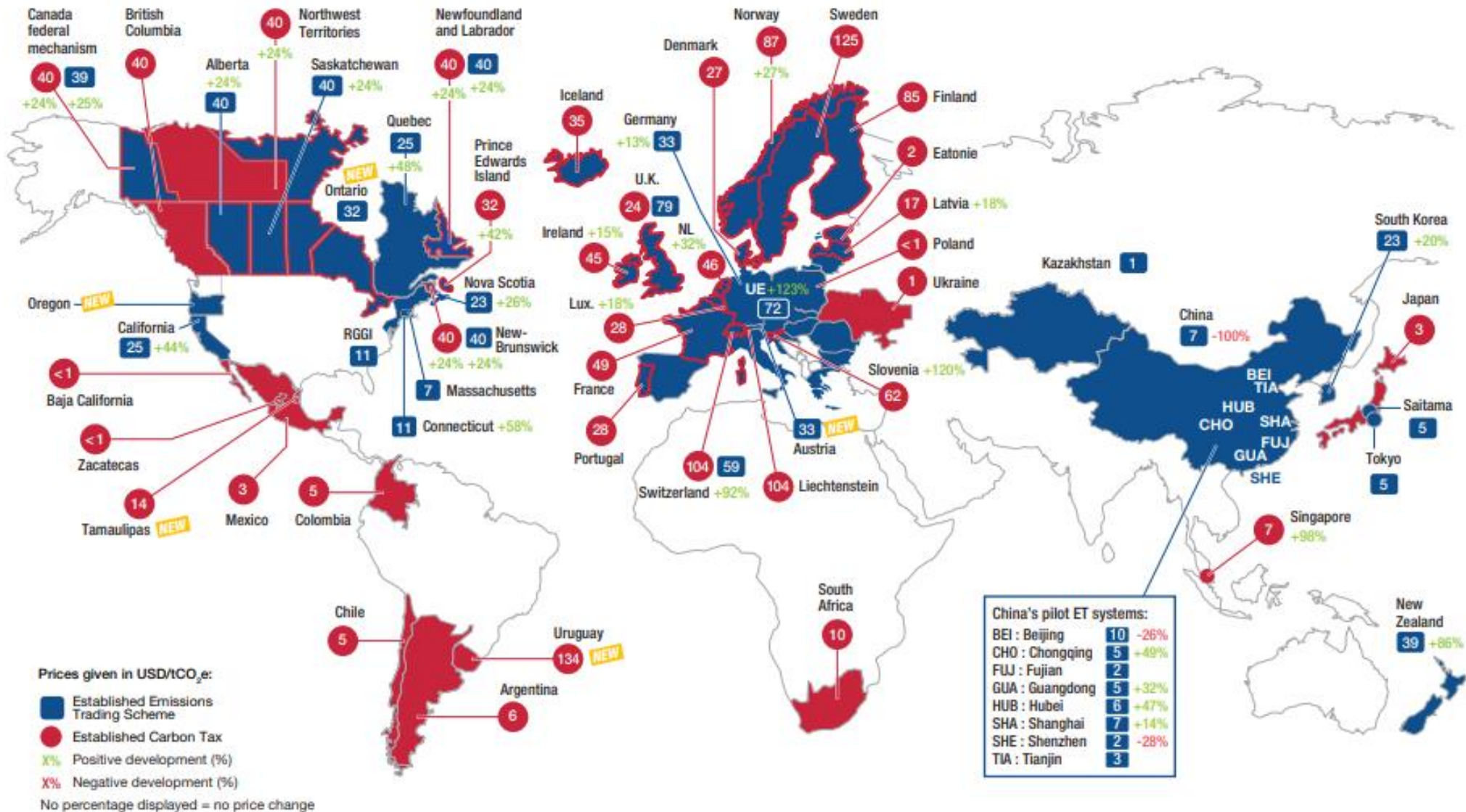
Two main pricing systems:

1. Carbon tax
2. Emissions trading system (ETS)

Underlying idea

- A high carbon tax impacts the creditworthiness of corporates
- This impact is different from one issuer to another one
- Identifying for each company the carbon price that would lead the default probability in the Merton model to exceed a certain threshold

Two forms of carbon pricing

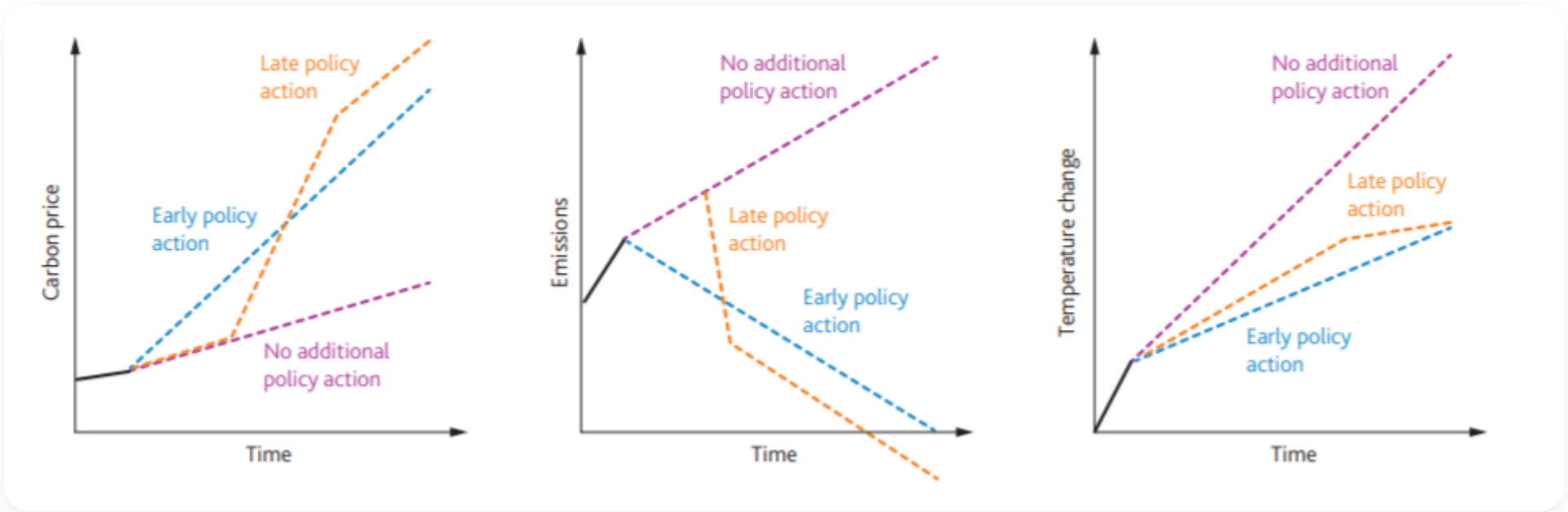


Audience Poll

Which of these scenarios do you think is most likely in real life?

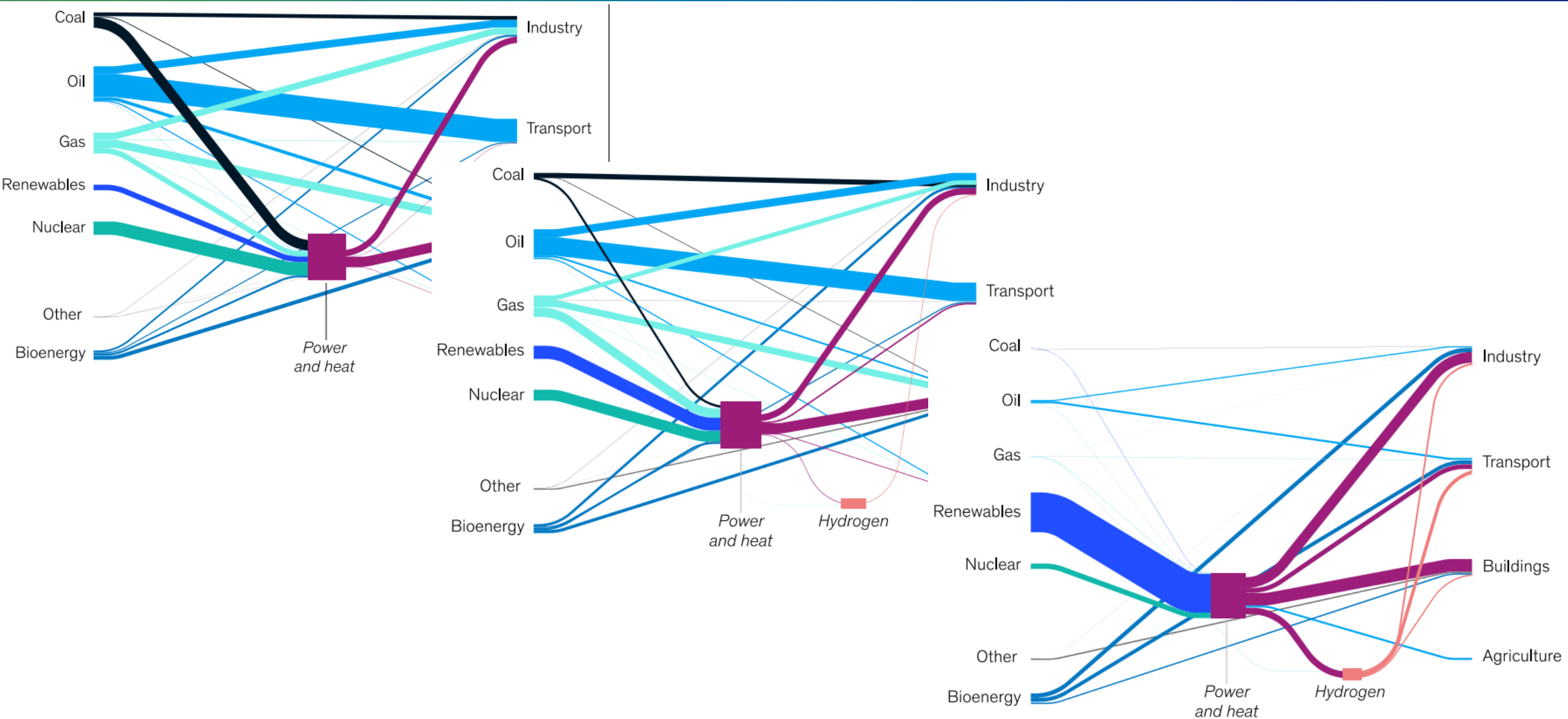
1. Early policy action: Orderly transition
2. Late policy action: Disorderly transition
3. No additional policy action: Hot house world

Illustrative pathways

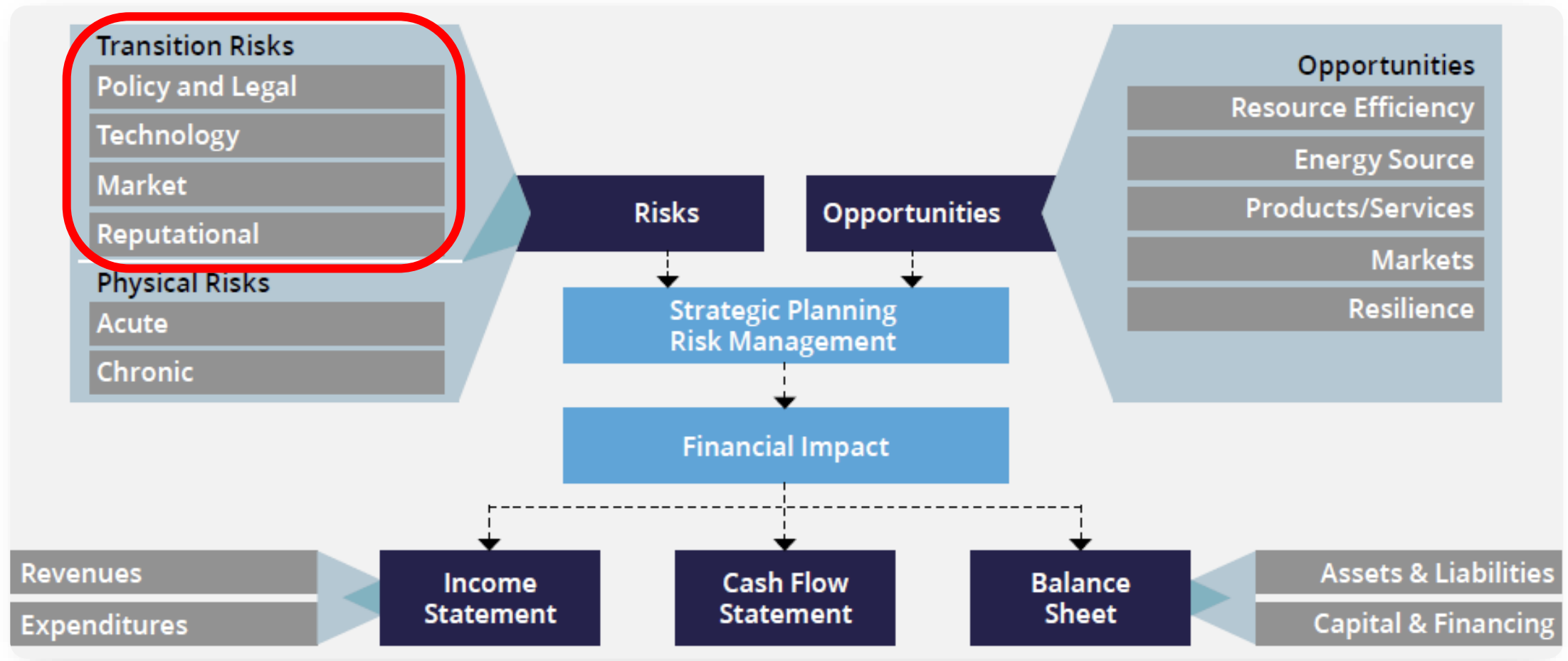


1. Early policy action: Orderly transition
2. Late policy action: Disorderly transition
3. No additional policy action: Hot house world

Transforming the global value chain into a net zero economy: 2017 / 2030 / 2050



Climate-Related Risks and Opportunities Create Financial Impacts



SUSTAINABLE FINANCE

Concept of sustainable finance (European Commission)

*“Sustainable finance refers to the process of taking **environmental, social and governance (ESG)** considerations into account when making investment decisions in the financial sector, leading to more long-term investments in sustainable economic activities and projects.*

***Environmental considerations** might include climate change mitigation and adaptation, as well as the environment more broadly, for instance the preservation of biodiversity, pollution prevention and the circular economy.*

***Social considerations** could refer to issues of inequality, inclusiveness, labour relations, investment in human capital and communities, as well as human rights issues.*

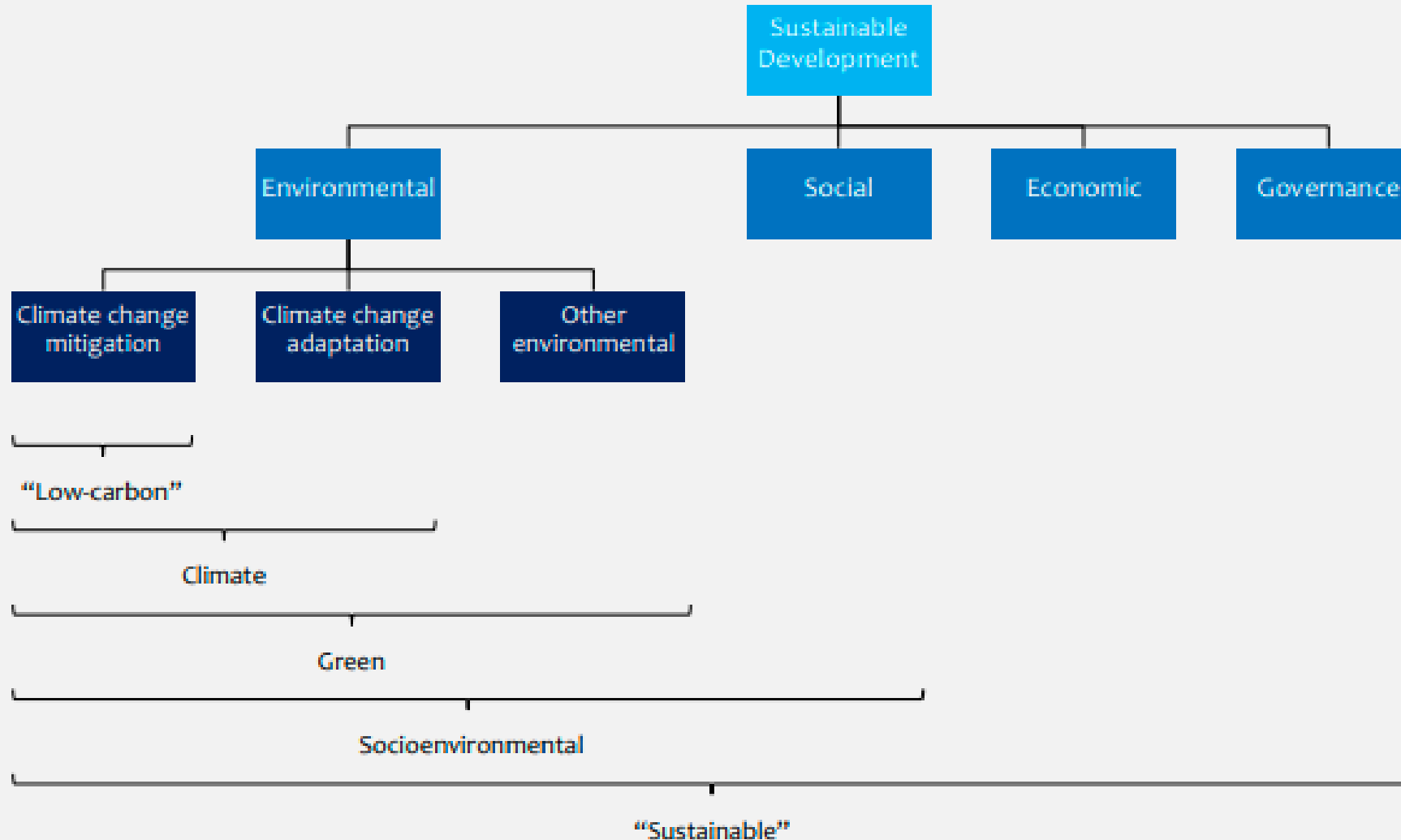
***The governance** of public and private institutions — including management structures, employee relations and executive remuneration — plays a fundamental role in ensuring the inclusion of social and environmental considerations in the decision-making process.”*

Many words, one concept

- **responsible investment (RI)** – first SRI mutual fund (Pioneer Fund) was created in 1928 – qualitative approach
- **CSR** - publication of “Social Responsibilities of the Businessman” by Bowen (1953).
- **ESG** – in 2005 UNEP - report about the legal use to integrate ESG issues by pension funds, insurance companies and asset managers - quantitative approach
- **sustainable finance**

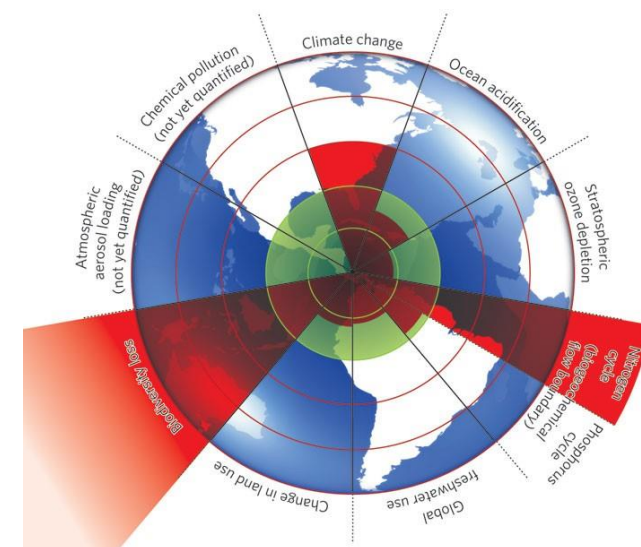
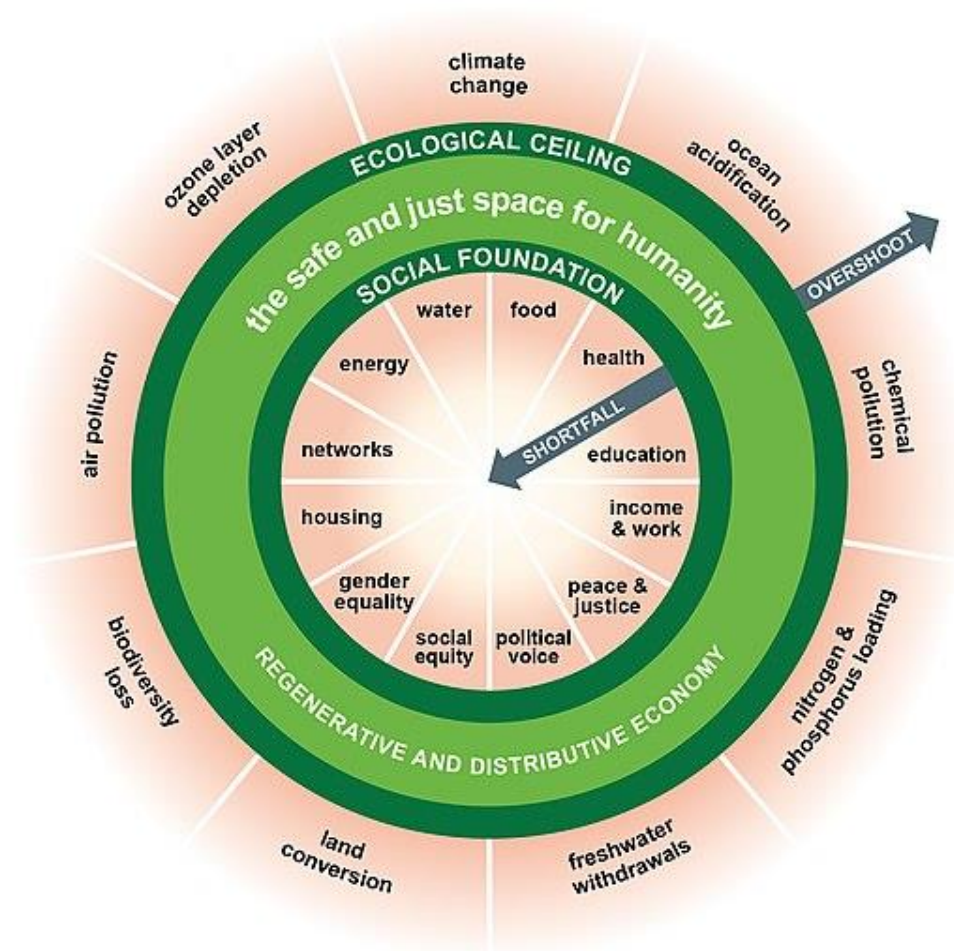


Defining Sustainable Finance



Source: UN (2018: 6).

Doughnut (economic model)



Source: Oxfam. Rio+20. Rockström et al (2009)

ESG ecosystem (countries)

- The World Bank database dedicated to sovereign ESG indicators is certainly the easiest way to understand the most common topics relevant to ESG analysis. The database is available at the following webpage: <https://datatopics.worldbank.org/esg> . It contains 67 ESG indicators grouped into 17 themes
- If we would like to focus on a small number:
 - E pillar - CCKP, EDGAR and Climate Watch
 - S pillar - FAO, ILO and WHO,
 - G pillar - the Worldwide Governance Indicators (WGI, <https://info.worldbank.org/governance/wgi>).

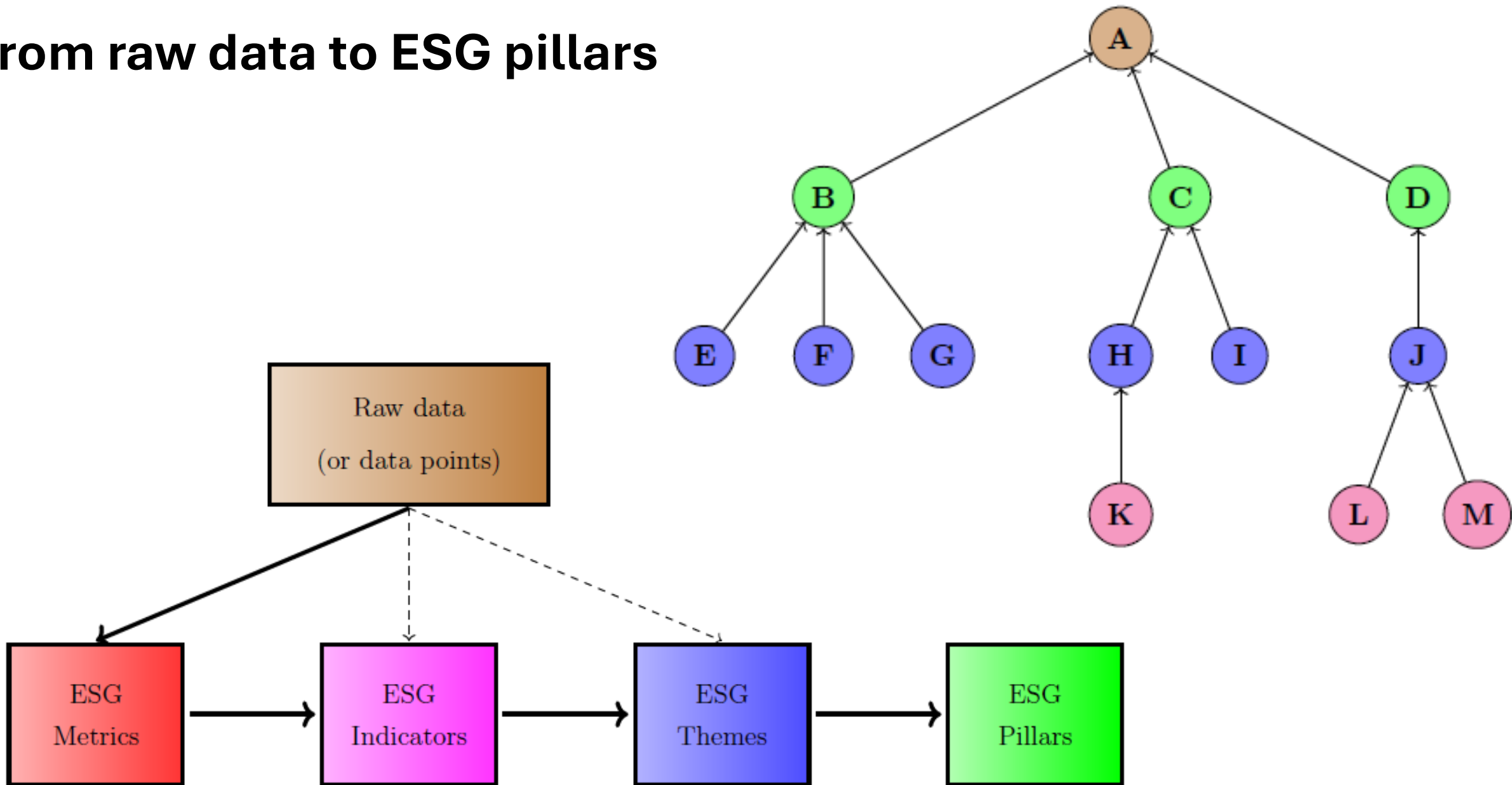
ESG ecosystem (companies)

There are 3 main challenges and barriers to corporate ESG data:

1. Data coverage (how to collect data for all the listed companies?);
2. Data sourcing (where to find the data?);
3. Data quality (what is the accuracy of the collected data?)

Contrary to sovereign issuers, raw data for corporate issuers are more difficult to find; they are not in open source data or they can only be manually collected (e.g., annual reporting). The ESG Data Cartography, <https://www.institutlouisbachelier.org/en/esg-data-cartography>, proposes a comprehensive list of ESG data with 140+ data sources. The user can filter the databases by accessibility (free, open source, partially free and proprietary).

From raw data to ESG pillars



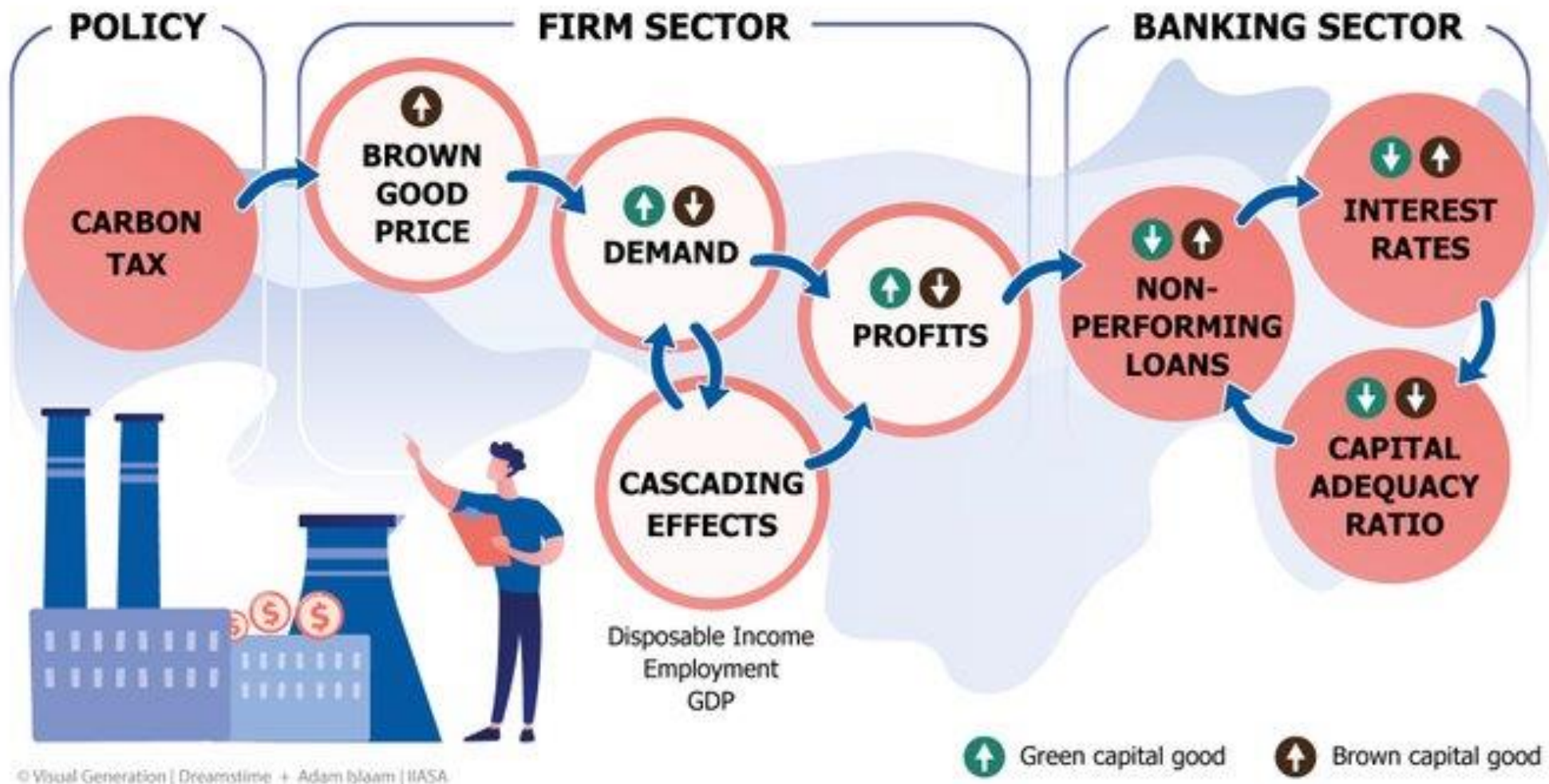
ESG and scoring theory

- ESG scores are generally the result of an unsupervised statistical method
- ESG scoring is more of an expert system than a true scoring model

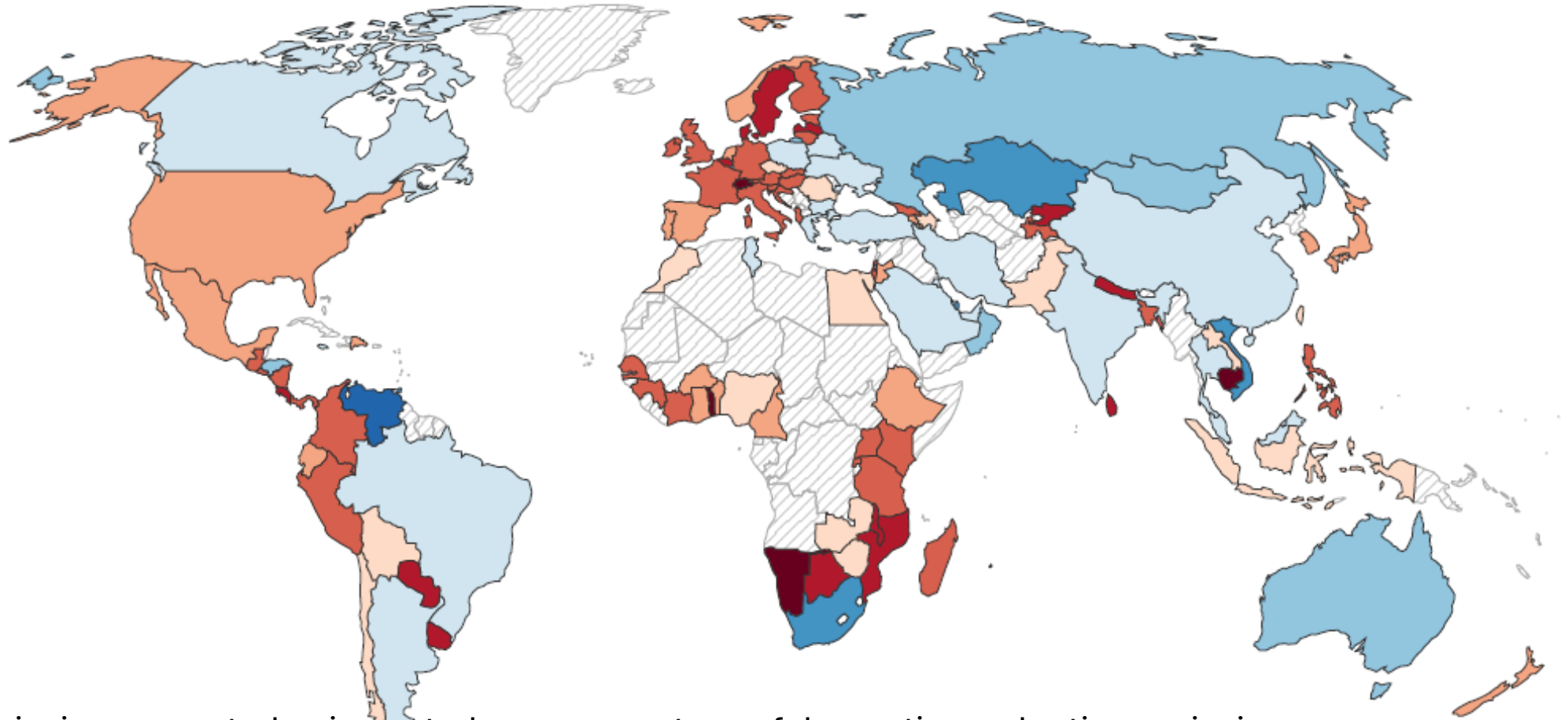
The fact that ESG scoring is designed as an unsupervised statistical process makes it difficult to assess the quality of ESG rating systems, particularly with respect to materiality

Most ESG scoring systems are sector neutral, meaning that the normalization is done at the sector (or industry) level, not at the issuer universe level. ESG scores are then **relative** scores (with respect to the sector/industry), **not absolute** scores.

What we expect

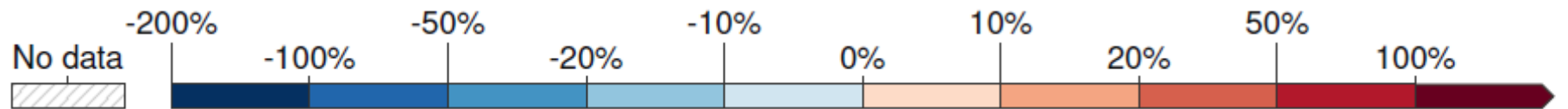


Carbon emissions embedded in trade, 2020



This is measured as emissions exported or imported as a percentage of domestic production emissions. Positive values (red) represent net importers of CO2. Negative values (blue) represent net exporters of CO2.

Source: <https://ourworldindata.org/consumption-based-co2>.



Carbon footprint

- is generally measured in carbon dioxide equivalent (CO_2e), which is a term for describing different GHGs in a common unit:

equivalent mass of CO_2 = mass of the gas x GWP of the gas

where the GWP of a gas is the amount of CO_2 that would warm the earth equally (expressed in kilogram):

- 1 kg of methane corresponds to 28 kg of CO_2 and
- 1 kg of nitrous oxide corresponds to 265 kg of CO_2 .

Carbon footprint

- total greenhouse gas (GHG) emissions caused by a given system, activity, company, country, or region. GHG are made up of water vapor (H₂O), carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), Ozone (O₃), etc.

$$m = \sum_{i=1}^n m_i \cdot GWP_i$$

where:

m - the mass of CO₂ equivalent,

m_i - the mass of the i -th gas

GWP_i - the global warming potential of the i -th gas ,

n - the number of gases.

Context of Sustainable finance

Sustainable finance is defined as the incorporation of environmental, social, and governance (ESG) principles into business decisions, economic development, and investment strategies.

Key Pillars	Key Themes
Environment	Climate change
	Natural resources
	Pollution and waste
	Opportunities and policy
Social	Human capital
	Product responsibility
	Relations
Governance	Corporate governance
	Corporate behavior

Green Asset Classes (GSS Market)

Asset Class	Type
Equity	Negative (exclusionary) and positive (best-in-class) approach
Debt Fixed Income	Traditional corporate bonds
	Traditional sovereign bonds
	ESG money market funds
	Green bonds
	Social bonds
	Sustainability bonds
Debt Bank Loans	Green mortgage-backed securities (MBS)
	Green loans
Alternative Investment	Sustainability-linked loans
	Green real estate investment trusts (REIT)
	Private equity (PE) and venture capital (VC)

Types of green (product snapshot)

Debt Fixed Income	Examples
Traditional corporate bonds	Bonds with proceeds used for funding new and existing projects with environmental benefits (not labeled).
Traditional sovereign bonds	Bonds issued by governments with proceeds used for funding projects with environmental benefits (not labeled).
Green bonds	Specific bonds that are labeled green, with proceeds used for funding new and existing projects with environmental benefits.
Social bonds	Bonds that raise funds for new and existing projects that create positive social outcomes.
Sustainability bonds	Bonds with proceeds that are used to finance or refinance a combination of green and social projects.
Green mortgage-backed securities (MBS)	Green MBS securitize numerous mortgages that go toward financing green properties (Fannie Mae)



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